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RESEARCH ARTICLE

REVISED Blockchain-enhanced electoral integrity: a robust model for secure digital voting systems in Oman

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Abstract

Background

Ensuring the security and trustworthiness of a digitized and automated electoral process remains a significant challenge in democratic systems. As digital voting systems are increasingly being investigated worldwide, ensuring the integrity of the process using robust security measures is of great importance. This paper presents a simplified model to enhance electoral integrity by leveraging Blockchain technology in the context of Oman's digital voting system. The model uses Blockchain technology to create a secure and trustworthy voting environment, addressing key vulnerabilities in digital electoral systems.

Methods

The research utilized a quantitative approach, employing an experimental design methodology using open-source software to simulate voting systems. Synthetic population data is used to operate these systems, while advanced biometric authentication technologies verify voter identities. Blockchain technology is leveraged to ensure secure vote recording, with smart contracts used to authenticate voters and securely record votes. Additionally, synchronous transactions are executed for both voter registration and voting

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processes, enhancing the overall security and efficiency of the system.

Results

The experimental results show that Blockchain enhances electoral integrity and security in Oman's voting system, improving elections' transparency and reliability. The performance evaluation of the model focuses on efficiency, reliability, and scalability metrics. Asynchronous transactions are utilized to improve processing time for voter registration and voting. Election administrators can manage, monitor, and certify election results, while Ethereum nodes ensure decentralized verification and transparency in the voting process.

Conclusion

This research offers insights for policymakers to consider Blockchain for electoral reforms, addressing issues like data integrity, fraud prevention, and transparency to boost voter trust. A strong regulatory framework and public awareness are crucial for successful implementation. Pilot projects are needed to assess Blockchain's practical impact. Oman could lead global innovation in electoral technology, though infrastructure and public resistance challenges must be managed.

Keywords

Blockchain Technology, Electoral Integrity, Digital Voting Systems, Voter Security, Transparent Elections, Secure Voter Authentication, Decentralized voting, Oman



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REVISED Amendments from Version 2

This new version has been revised based on the reviewers' feedback, a point by point revision have been made to address the reviewers comments. The revised manuscript includes an extended literature review on Ethereum scalability and smart contract security. Authors clarified the differentiation between experimentally supported conclusions and speculative statements regarding infrastructure readiness and public trust. New sections were added to compare the model with existing solutions and highlight its unique suitability to the Omani context, detailing its practical, decentralized design and integration with Oman's e-governance. Additionally, the 'Results and Discussion' section was expanded to include a theoretical analysis of integrating Layer 2 (L2) solutions for scalability and outlined plans for expanded threat modeling and pilot studies with real-world stress-testing.

Any further responses from the reviewers can be found at the end of the article

1. Introduction

Democracy is one of the most essential pillars of a democratic country, where the selection of the Government is made through elections and voting by the country's citizens.¹ The vote can be considered a formal and documented way of choosing a candidate of choice from the multiple contestants during the election process. This process typically involves using a ballot, a show of hands, or other methods to indicate one's preference.² The security and accessibility of the voting process have higher priority as the elections are directly connected to the effectiveness of the democracies and a better future of the citizens of the country. Ensuring that every qualified individual can participate in the voting is vital to preserving the integrity and legitimacy of democratic systems.³ It ensures that all perspectives are considered, strengthens equity and justice, and improves public confidence in the electoral process.⁴ Historically, voters have used hand gestures to signal their choice with a thumbs-up for approval or a thumbs-down for disapproval.⁵ The candidate who received the highest thumbs-up was declared the winner. This straightforward method of voting allowed for open and immediate visible feedback, making it easy to determine the outcome.⁶ This simple approach was effective in small gatherings or assemblies where decisions needed to be made quickly and transparently.⁷ The problems with the traditional electoral system are safety and security; handling the different interferences, ensuring legitimate users' access to the voting system, and maintaining the legitimacy of votes become challenging.⁸ Bracamonte et al. assert that a successful voting system meets the electoral system criteria that optimize the system's advantages for the citizens and the administration.⁹

There are several challenges associated with traditional and voting methods. Firstly, security concerns are significant, as these methods are susceptible to manipulation, fraud, and cyber attacks, raising doubts about the integrity of election results.¹⁰ Secondly, traditional voting lacks transparency, leading to questions about the results' legitimacy due to opaque processes. In addition, accessibility remains an issue for certain demographics, such as individuals with disabilities or those in remote locations, who may face barriers to participating in the voting process.¹¹ In addition, the high costs associated with physical infrastructure, printed ballots, and manpower contribute to the cost of elections. In addition, manual or electronic vote-counting processes can be time-consuming, resulting in delays in announcing results. Finally, inconvenient polling locations, long waiting times, or limited voting hours can discourage voter turnout, undermining democracy.¹²⁻¹⁴ Addressing these challenges is crucial to improving the efficiency, security, and inclusion of elections, and Blockchain technology offers potential solutions to many of these issues.

Blockchain technology comprises exchanging and storing verified transactions over a distributed network of computing nodes. The technology leverages techniques that ensure the integrity and security of data by maintaining a distributed ledger, also known as a Blockchain, which is a series of interlinked data records. Once added to the Blockchain, each transaction or record is linked to the previous entry using the one-way hash functions of the previous block, forming a chronological chain of blocks.¹⁵ The Blockchains are maintained across a network of computers, also known as qualifying nodes.

The decentralized verification of transactions before they are added to the tamper-proof Blockchain eliminates the need for a central authority, as all participants in the network have access to the same information, ensuring transparency and reducing the risk of fraud such as double-spending.¹⁶ The technology involves complex inter-process communication that verifies and validates each transaction, making it nearly impossible to alter or manipulate once the transaction is added to the Blockchain.¹⁷ For example, Bitcoin, created by Satoshi Nakamoto in 2008, is a popular application of Blockchain technology that facilitates the payments of transactions between two parties without any intermediary, such as credit card companies and banks.¹⁷

The proposed electronic voting model leverages Blockchain technology to enhance security, transparency, and trust in the electoral process. This research considers the case of Oman to present and compare both traditional and electronic

systems in the election process, which is reflected in the electronic voting system used in the recent Omani elections held on 27th October 2019. This research considers the case of the electoral system in Oman to present and compare traditional and electronic electoral systems. The contributions of this research are as follows:

- Identify and discuss security and trust issues with traditional electoral systems.
- To propose and prototype a simplified Blockchain-based electoral system.
- To implement and test the smart contracts required for Blockchain-based electoral system.
- Evaluate the proposed digital Blockchain-based electoral system regarding performance matrices such as the time required for registration and voting.

The rest of the paper is organized as [Section 2](#) discusses the preliminaries, [Section 3](#) presents related works published on the Blockchain and electronic voting, and [Section 4](#) contains system models and methods used in the proposed model. [Section 5](#) presents the experimental setup of the study while [Section 6](#) presents a detailed discussion of the results followed by [Section 7](#) conclusion.

2. Preliminaries

2.1 The Omani Election System: An overview

In Oman, two primary council bodies form the “Majlis Oman”: the Majlis Al-Shura (or Lower House) and the Majlis Al-Dawla (or Upper House). The members of the Majlis Al-Dawla are chosen by the reigning Sultan, with Sultan Haitham continuing the practice established by his predecessor, Sultan Qaboos. These members are either former government officials or influential citizens of the country.¹⁸ The general public selects the members of the Majlis Al-Shura through an optional voting process. The dual chamber system balances the leaders’ public representation and experience to maintain political stability. Sultan has ultimate power for the final approval, but on the other hand, Majlis Oman, with sufficient powers, has the authority to review and approve new laws and budgets.

The Majlis Al-Shura of Oman, a consultative assembly, was formally established in 1991, succeeding the nation’s traditional consultative council.¹⁹ This legislative body comprises delegates who serve as representatives for each of Oman’s 61 administrative governorates, known as **Wilayats**. The apportionment of representation within the Majlis Al-Shura relies on the demographic size of each Wilayat. A single representative for each Wilayat, but in cases where a Wilayat’s population exceeds 30,000 residents, is afforded a dual representation. The tenure for each member of the Majlis Al-Shura is a fixed term of four years.²⁰

The Ministry of Interior (MoI) of Oman is the governing authority responsible for administering and overseeing the country’s electoral process. To be enfranchised and exercise their right to vote, prospective voters are required to first register with the MoI. This registration process is a prerequisite for inclusion in the official electoral rolls. Furthermore, several specific criteria must be met for an individual to be deemed eligible to vote.²¹

- They must be a citizen of Oman.
- They must be at least 21 years old.
- They must cast their vote in either their Wilayat (administrative division) of residence or the location listed on their Omani ID card.

The voting process in Oman requires voters to present themselves at a designated election station within their Wilayat on the day of the election. At the station, an administrative committee manually verifies each individual’s identity using their official ID card. Upon successful verification, the voter is issued a ballot paper. The voter then confidentially marks their preferred candidate and deposits the ballot into the secure selection box. It is a strict rule that voters can cast only one vote for a single candidate. Following the election’s conclusion, an audit committee at each station manually reviews and tallies all ballots to ascertain the final election results. This extensive, labor-intensive procedure is associated with various drawbacks, including:

- Casting a vote requires a physical presence at an election station, which can be inconvenient for voters, as it necessitates travel to their respective Wilayats.

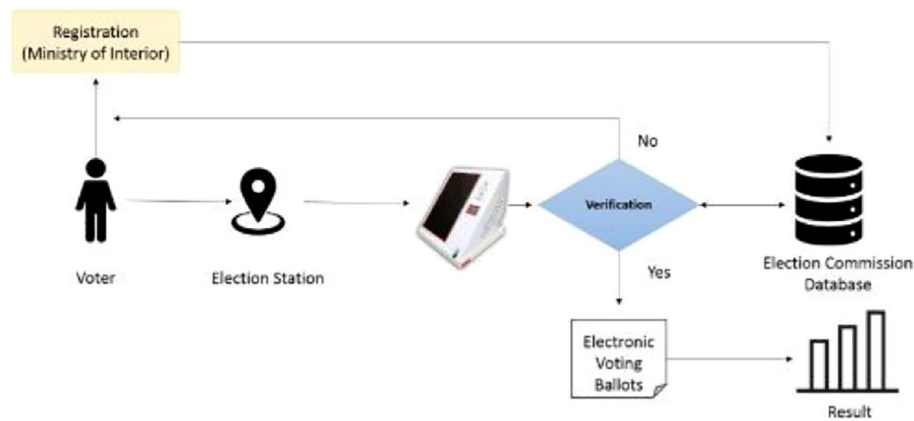


Figure 1. Use case model of the system.

- The manual counting of ballots is time-consuming and vulnerable to various issues, such as human error, which can lead to miscalculations and even intentional manipulation.
- The lack of automated or digital verification in the vote-counting process creates a significant opportunity for fraudulent activity.

Oman has significantly changed to address the challenges of the traditional, paper-based voting system. The country has stopped using paper ballots and implemented a modern electronic voting system. This technological shift enhanced the electoral process by making it more traceable and verifiable, thereby increasing security and public trust in the election results.

The ninth general election in Oman, which took place on October 27, 2019, was conducted using an electronic voting system. This electronic electoral process utilized a device called “Sawtak,” deployed at every polling station. The device automates the verification process; once a voter enters their ID number, the system automatically authenticates their identity, followed by a biometric fingerprint scan to confirm their eligibility. Upon successful verification, an electronic ballot listing the candidates is displayed on the device for 120 seconds, allowing the voter to select. This ballot is only accessible to authorized individuals; it is not available to those not registered with the Ministry of Interior (MoI) or to individuals attempting to vote at an incorrect polling station. After a voter has selected their candidate, the device generates and displays an electronic ticket. This ticket, which includes the voter’s name and their chosen candidate, serves as a digital receipt to confirm the successful completion of the voting process. [Figure 1](#) clearly illustrates the election process in Oman, its various stages from voter registration to the final tallying of results.⁵

2.2 Blockchain

A Blockchain is a series of data blocks arranged in a specific order.²² Each new block is cryptographically linked to the one before, forming a secure chain. The cryptographic reference, also known as block hash, is usually the secure hash value of the contents in the previous block. If M are data contained in a block, then $\delta = H(M)$ is the block hash, where $H(\cdot)$ is a secure, one-way function such as SHA-2, SHA-3, among others. A block can contain different information such as block index, timestamp, previous block’s block hash, number of transactions in the block, and the list of transactions. The block size reflects the capacity of the entire block and the number of transactions it can carry. The initial block of the Blockchain is called a genesis block with a block hash of δ_0 . As shown in [Figure 2](#), blocks $B_0, B_1, \dots, B_N$ are the data blocks forming a Blockchain.^{8,23} Thus, a block in a Blockchain contains multiple transactions, and a transaction contains the data values involved in the transaction.⁸

The power of a Blockchain is born out of the feature that if any data in a block is to be changed, then every subsequent data block should be modified accordingly because of the change in the block hash. This property aids in ensuring the integrity of the data blocks if the cryptographic reference to the recent block is known or agreed upon.²⁴ For instance, say the block hash of the recent block B_N is known to be δ_n . If data in block B_1 is altered, its block hash is altered, eventually altering the block hash of B_N . Hence, Blockchain securely stores data to prevent alteration once recorded unless a unanimous agreement exists among all parties involved.²⁵ In other words, the property of data immutability is achieved through a consensus mechanism that ensures all transactions are verified and added to the Blockchain transparently and irreversibly. Blockchain provides a trustworthy ledger where data integrity and transaction history are preserved, fostering transparency and reliability in various applications such as supply chain management, financial transactions, and voting

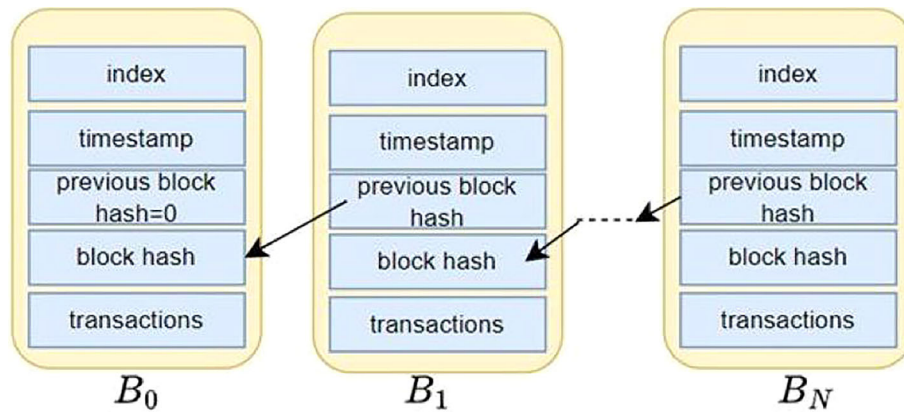


Figure 2. Structure of a simple Blockchain ledger.²² This figure has been reproduced with permission from Adhikari et al. (2023).²²

systems (i.e., the approval of above 50% of the network nodes is required).²⁶ De Filippi and Hassan²⁶ consider Blockchain as a decentralized database (or state machine) that considers cryptographic primitives for the surety of data integrity and authenticity. It has also been known as electronic and decentralized, sometimes as an immutable transaction ledger that provides cryptographic verification. These definitions underscore the fundamental aspects of Blockchain technology, emphasizing its decentralized nature, cryptographic security measures, and the immutability of transaction records.²³

2.3 Blockchain Network

One of the important aspects of a Blockchain ledger is what makes a valid data block and who can add a new one. A mechanism to add only a verified data block to the chain can aid in maintaining the perpetual integrity of the data.²⁴ This security feature is provided by sharing the Blockchain among a special group of computing nodes known as a Blockchain network. The broadcast network of computing nodes is known as a Blockchain network (BCN).²⁴ It can maintain a copy of a synchronized storage, which is defined as a Blockchain. Each node in a Blockchain network runs a protocol for maintaining data consistency/integrity, verifying a transaction before it is added to the Blockchain, maintaining operation transparency, and maintaining privacy through anonymity. A Blockchain protocol permits anyone to propagate transactions through the network for verification and to be recorded in the ledger. Some special nodes, known as miners, verify and add well-formed transactions to the ledger. The miners in the network use consensus algorithms, such as Proof of Work (PoW), and Proof of State (PoS), to agree upon the block to be added to the Blockchain.^{23,27} Since multiple verifiers verify the validity of a transaction, Blockchain technology (Blockchain and Blockchain network) offers decentralization and disinter-mediation of transactions between users internationally.²⁸ In brief, Blockchain technology provides a trustworthy ledger where data integrity and transaction history are preserved, fostering transparency and reliability in various applications such as financial transactions, supply chain management, and voting systems (i.e., the approval of above 50% of the network nodes is required).²⁶

Figure 3 a simple Blockchain network with three network nodes n_1 , n_2 , and n_3 . Each node in a Blockchain network maintains a copy of the underlying Blockchain ledger $B_0 \leftarrow B_1 \leftarrow \dots \leftarrow B_j \leftarrow B_n$. If a transaction T_j is sent to node n_3

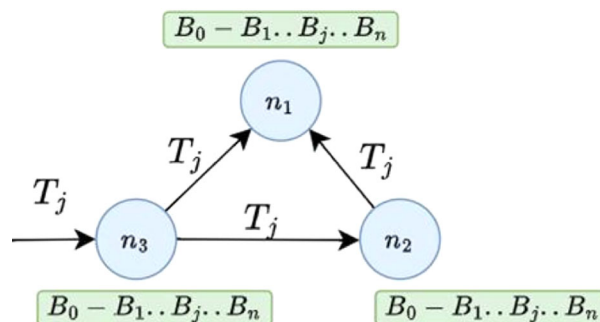


Figure 3. A simple network of Blockchain with three nodes.²² This figure has been reproduced with permission from Adhikari et al. (2023).²²

for validation and to be recorded in the existing Blockchain, it is broadcast to connected nodes, which will also verify its correctness.²² Many authors^{23,27} agree that Blockchain allows anyone worldwide to send or participate in transactions, providing greater assurance that they are engaging with the rightful owner. This immutability is achieved through a consensus mechanism that ensures all transactions are verified and added to the Blockchain transparently and irreversibly.

2.4 Blockchain with Electoral Integrity from a Governance and Democracy Perspective

Blockchain technology has captured their attention as scholars and policymakers seek to bolster integrity in elections. Dominating the conversations was the impact of digital technology on institutional trust and its relation to electoral systems and the confidence voters placed in those systems.²⁹ The available literature reveals that implementing Blockchain in voting systems has advantages and challenges.

Defenders of Blockchain note that its features, particularly decentralization and the inability to alter information, can remarkably enhance the transparency of elections. Ohize et al.³⁰ stress that mobile voting systems based on Blockchain technology can potentially eliminate voter fraud and increase turnout by providing a secure and verifiable system for voting. They highlight that Blockchain-based voting systems can be audited by anybody, enhancing credibility and faith in the system and electoral framework. Regardless of these advantages, Assouline et al.³¹ argue that some privacy issues, cybersecurity, digital skills, and the exclusion of certain demographics add to the criticism of using Blockchain technology in elections. There needs to be sufficient public awareness and development of the relevant technology to enable everyone's participation.

Similarly, Farooq et al.³² explain how Blockchain can help protect voter registration information by making alterations transparent and traceable. Jafar et al.³³ discussed that electronic voting machines with Blockchain technology could issue a printed paper receipt to the voters, which would aid in audits and boost trust in the vote-counting process. Moreover, Olaniyi et al.³⁴ underscore the risks of applying new technologies to the election process. They contend that elections are critical undertakings wherein things cannot go wrong, since poorly applied Blockchain systems would lose public trust. Implementing any technological application should be preceded by robust public awareness campaigns managed by reputable, non-partisan election officials.

The use of Blockchain in the 2023 elections in Guatemala is an example that illustrates the expected results of the implementation. As stated in a Democracy and Society³⁵ report, the technology was applied to restore trust in the electoral system after allegations of fraud in preceding elections. Storing each transaction in a ledger using Blockchain technology boosted the electoral system by easily detecting attempted manipulations.

One of the distinguishing features of democracy is transparency and digital democracy. As noted by many researchers, Blockchain technology requires a guarded approach regarding security and public trust.³⁶ These two elements demonstrate the utmost necessity in maintaining the public's trust in the system. Transparency of information coupled with accessibility indicates an open, robust form of Government, which can be achieved through Internet technology. With innovative ideas purchasable through the Internet, civic engagement engages us to welcome and engage in democracy through active voting. Voting becomes a civic duty for citizens of democratic countries as it directly influences their choices and decisions.

In this regard, elections are a striking example of implementing civic engagement. However, they tend to ignore that civic engagement can be defined in a multi-dimensional framework that features systems thinking that captures the complexity of social systems, making a participatory approach more appealing. Regardless of who utilizes the notion of digital democracy alongside open Government, the digitally achieved, or instead opened democracy acts as a form of power that the Government can exercise and aid in refining the understanding of democracy through freedom.

2.4.1 Theoretical Foundations and Potential Benefits

Blockchain functions as a transparent public ledger incapable of modification, which opens the possibility for a new method of ensuring fair elections. Employing specific coding techniques, voting systems based on Blockchain technology can offer End-to-End Verifiable (E2E) elections.³⁷ E2E implies that votes are cast on time, midstream, stored, and counted while stored. This capability can resolve problems associated with election fraud, risk of hacking, and personnel error. However, adding Blockchain does not automatically provide more trust in elections.³⁸ Rather, the trust in Blockchain voting is founded on the perception of voters, political leaders, and election officers. Without trust and knowledge of such systems, citizens cannot endorse democracy.

2.4.2 Trust in Institutions and Electoral Legitimacy

Trust in institutions is one of the central issues in political science. To what degree do people consider that political institutions are executing their roles properly? Standard voting systems, dependent on institutions like election commissions, political parties, and independent monitors for free and fair elections, rely on multiple functions working synergistically to acquire an equilibrium bloc.³⁹ It's more a question of voters' trust in the technology than in existing institutions and whether Blockchain technology, which offers a mechanism without a central authority, will make freer elections possible.

Most studies indicate a general decline in confidence in voting systems across the globe. People are apprehensive regarding cheating, vote tampering, and cybersecurity threats.^{40–42} Blockchain is suggested as an answer to this problem because there is no need to trust a central authority. All the transactions (votes) can be independently verified. Political analysts say elections are primarily social and not purely technical.⁴³ If people remain doubtful about Blockchain security, ease of use, and the possibility of government tampering, increased reliability of elections will not be guaranteed.

Research indicates mixed results on users' trust in voting via the Internet. Some suggest that younger, more adept technology users will trust elections run on Blockchains while older users grapple with the system's clarity and transparency. Moreover, in cases where there is extreme political division within parties, new election platforms are likely to be more divisive than unifying.

2.4.3 Voter Perceptions and Digital Transparency

Public trust in the fairness and accessibility of the electoral processes is fundamental to their credibility. A Blockchain framework could worsen fears and misunderstandings without clarity, education, and transparency.

Perceived Opacity vs. Actual Transparency

Although Blockchain can improve transparency by allowing voting processes to be publicly verified, most voters lack the technical skills necessary to audit Blockchain systems.^{44,45} Without straightforward verification mechanisms, Blockchain implementation could inadvertently generate a black-box scenario where the community must depend on trusted specialists to navigate the opaque system instead of analyzing the results independently.

Digital Literacy and Accessibility

The positive impact of Blockchain technology on electoral integrity relies on the assumption that all voters have the requisite digital literacy skills to engage with a Blockchain voting system. Nonetheless, some studies indicate that the digital divides informed by education, age, and socioeconomic status can extend existing disenfranchisement.⁴⁶ That said, implementing Blockchain voting needs to be preceded by efforts to educate voters about the technology so that they have an adequate understanding and trust in the system.

Role of Social Media and Disinformation

The capability of Blockchain technology to secure electoral data does not address the challenges of misinformation. Today's elections are as much about the social media narratives, spin, and misinformation surrounding them as they are about casting votes.⁴⁷ Even if there were a voting system based on Blockchain that was utterly impenetrable, trust in the system would still be undermined in highly polarised elections due to rampant inaccurate claims regarding the results.

2.4.4 The Role of Digital Transparency in Democratic Legitimacy

Democratic legitimacy defines whether a government has the right to rule and if it acts with the consent of the citizens. Improving digital transparency with Blockchain technology may help in improving legitimacy by:

- Offering tangible proof that hacking did not occur with the voting.
- Minimizing dependence on authoritative sides reduces the chance of bias from the institutions.
- Increasing the possibility of international observation of elections, allowing outsiders to witness the processes live.

Elections conducted using Blockchain technology may expose previously undetected discrepancies in the voting process. These discrepancies, however, could be misinterpreted as fraudulent activity, thus eroding trust in the electoral system. While Blockchain may reinforce election integrity, its effectiveness will hinge on public perception, institutional support, and integration into current electoral systems.⁴⁸ Should insufficient public confidence and trust exist, an inclusive strategy could lead to opposition to the elections. Lastly, to bolster the democratic image of a nation, issues surrounding democratic election integrity need comprehensive public trust initiatives and civic education campaigns that examine governance issues, establish trust in the system, and systematically approach the integration of Blockchain technology.

3. Related work

The first electronic voting system was conceptualized by “David Shaum” in the early eighties, where public-key cryptography was utilized to ensure that electors were selected anonymously and to separate the voters’ identity from the ballots.⁴⁹ Estonia has adopted the electronic voting system for its national election in 2005.^{50,51} However, Denmark’s Liberal Alliance in 2014, started planning to integrate the Blockchain to enhance the privacy and security of national elections in Denmark.⁸

During the later months of 2018, South Korea’s National Election Commission introduced a trial of a Blockchain-based online voting system for the South Korean elections, aiming to enhance the reliability and security of the voting system. As a cutting-edge innovation rooted in robust cryptographic principles, Blockchain technology facilitates easier data access for candidates and observers, contributing to greater transparency and trust in the electoral process. However, they believe that applying Blockchain-based voting will raise the rate of buying votes. As explained in the previous section, Blockchain technology can validate whether a bribed voter adhered to their part of the deal or not.⁵² Overall, the trial of using Blockchain-based technology in electronic voting has been successful.

This foundation empowers Blockchain-based applications to capitalize on advanced cryptographic techniques, enabling them to implement highly adaptable security solutions. By leveraging these cryptographic abilities, Blockchain enhances the security of data and transactions and ensures transparency and integrity across decentralized networks.¹ Nearly all discussions about Blockchain converge on a shared concept, though articulated with variations.

Experts believe the traditional voting system has problems like the security of the votes; hence, Blockchain is the perfect solution for securing elections. Hence, technologists consider it an ideal platform for global democratic systems.¹³ Blockchain technology in the electronic voting system protects the secrecy of the ballots and is ubiquitous. Further, the free and open source peer review software allows a free and independent audit of the result and decreases the confidence level required from the organization or the election entity.⁵³

The distributed ledger technology allows multiple points to have the ledger instead of restricting to one location; this means that all points can have the same results, unlike in traditional ballot boxes.¹³ The decentralized structure of Blockchain reduces the chance of failure or any records getting corrupted. Different transactions have blocks holding previous electors’ transactions to ensure the voting validity, security, and connectivity. If the chain has been compromised and altered, it can be easily detected by observing the broken link between blocks. In the real-time Blockchain-based voting system, once the voter casts a vote, it is sent to one of the nodes and later added to the Blockchain. The voting system has a node in each area to maintain decentralization of the democratic framework.⁸ The system distributes power and control, and prevents any single person or group from manipulating the election and strengthening the democratic process.¹

The consensus algorithm makes the Blockchain dominant over other technologies, and hence, it has increased the adaptability of the Blockchain as it can perform tasks faster, more accurately, and more efficiently than bureaucratic systems based on paper voting.^{13,54}

As technology advanced, many countries began integrating Blockchain technology into their voting systems to enhance transparency, privacy, and accessibility in the voting process.^{55,56} Furthermore, many countries adopt Blockchain technology in their e-government systems.⁵⁷ However, the integrity of the voting process hinges on a consortium of trusted validators tasked with either approving or rejecting transactions. A recent study by Ref. 58 proposed an innovative adjustable Blockchain that can enhance the voting procedure and mechanism, and address security concerns. Their proposed framework offers an enhanced and more secure electronic voting process, ensuring data integrity through advanced hashing technology. Blockchain technology facilitates easier data access for candidates and observers, contributing to greater transparency and trust in the electoral process.⁵²

Blockchain technology is increasingly recognized as a cutting-edge innovation rooted in robust cryptographic principles. This foundation empowers Blockchain-based applications to capitalize on advanced cryptographic techniques, enabling

them to implement highly adaptable security solutions. By leveraging these cryptographic abilities, Blockchain enhances the security of data and transactions and ensures transparency and integrity across decentralized networks.⁵⁹ Nearly all discussions about Blockchain converge on a shared concept, though articulated with variations. According to the paper by De Filippi and Hassan,²⁶ it is described as a “decentralized database (or state machine) that relies on a set of cryptographic primitives to ensure data integrity and authenticity.” A digital ledger, or Blockchain, is decentralized and permanently secured with cryptography. These definitions underscore the fundamental aspects of Blockchain technology, emphasizing its decentralized nature, cryptographic security measures, and the immutability of transaction records.²³

Many authors agree that^{23,27} Blockchain allows anyone worldwide to send or participate in transactions, providing greater assurance that they are engaging with the rightful owner. The Blockchain securely stores data to prevent alteration once recorded unless a unanimous agreement exists among all parties involved.²⁵ This immutability is achieved through a consensus mechanism that ensures all transactions are verified and added to the Blockchain transparently and irreversibly. As a result, Blockchain provides a trustworthy ledger where data integrity and transaction history are preserved, fostering transparency and reliability in various applications such as financial transactions, supply chain management, and voting systems.²⁶ Blockchain technology offers decentralization and disinter-mediation of transactions between users internationally.²⁸ For a voting system utilizing Blockchain, real voting takes place in a Blockchain network. The votes by voters get sent to one of Blockchain’s nodes on the voting system, which then adds the vote to the Blockchain. The voting system has a node in each area to ensure the decentralization of the democratic framework.⁸ Blockchain technology safeguards election records from illegal access, alteration, and tampering with ballots.¹ This technology remains prominent due to the consensus algorithm that makes it difficult to vote on transactions and makes it difficult to alter or forge.⁵⁴

Using immutable storage, you can securely manage and maintain outsourced computational ledgers, protecting them against tampering and unauthorized access.⁶⁰ The work integrates Blockchain and machine learning for secure data transmission, employs NuCypher proxy re-encryption for efficient neighborhood encryption without cipher conversion, and utilizes ANN to optimize data delivery and record management in smart cities.⁶¹ A Blockchain framework enabled by Hyperledger Sawtooth, offering a secure and trusted execution environment where service delivery mechanisms and protocols ensure immutable ledger storage security, along with peer-to-peer network communication for both on-chain and off-chain industrial activities.⁶² This study proposes a lightweight Plenum consensus algorithm (BLPCA) for consortium Blockchains on Hyperledger Indy, leveraging optimized Byzantine Fault Tolerance to manage large-scale decentralized socioeconomic tax traffic within hierarchical systems efficiently.⁶³ The study highlights the benefits of a permissioned private Blockchain, addresses the challenges of the Proof-of-Elapsed Time (PoET) consensus mechanism in distributed applications, and proposes a lightweight middleware consensus called “B-LPoET” to improve Blockchain adaptability for private chains.⁶⁴ In various domains, Blockchain plays a critical role; hence, it has potential in the voting system. However, there are limitations in the traditional voting systems that need to be addressed by the secure voting system based on Blockchain.

3.1 Ethereum scalability solutions and smart contract security

Ethereum, launched in 2015, fundamentally transformed the Blockchain landscape by introducing smart contracts, laying the groundwork for decentralized applications (dApps) and the burgeoning decentralized finance (DeFi) ecosystem.⁶⁵ In 2022, Ethereum underwent a significant and pivotal upgrade known as “The Merge,” transitioning its consensus mechanism from Proof-of-Work (PoW) to Proof-of-Stake (PoS). This change was a monumental step in the network’s evolution, as it led to a dramatic reduction in energy consumption by more than 99% and laid the groundwork for future improvements in network scalability.⁶⁶ Ethereum’s transition to a Proof-of-Stake (PoS) consensus mechanism was a fundamental architectural shift, moving beyond simple energy efficiency. This strategic reorientation laid the essential groundwork for key scalability solutions like sharding and developing Layer 2 solutions. By making the base layer more efficient, PoS specifically enhanced its ability to support Layer 2 data requirements, positioning Ethereum for a multi-layered scaling approach.⁶⁷

Despite its popularity, Ethereum’s limited transaction speed and high fees, especially during peak use, have hindered its mass adoption. In Blockchain, scalability efficiently handles more transactions and users while maintaining security and decentralization. Smart contracts are vulnerable to security flaws that can lead to significant financial losses.⁶⁸ Their immutable and transparent nature permanently exposes any vulnerabilities to hackers, making them a prime target, with billions lost annually to exploits. Ethereum’s architecture is shaped by the Blockchain trilemma, which includes balancing decentralization, security, and scalability.⁶⁹ The tasks are divided between Layer 1, which handles the main network and focuses on security and decentralization. The Layer 2 solutions are built on top of Layer 1 to handle high transaction volumes, allowing Ethereum to scale without compromising its foundational principles.

3.2 Overview of Layer 2 solutions and their transformative impact

Ethereum can often feel slow; to overcome this, layer 2 solutions have become so important. Layer 2 speeds up the transactions and makes decentralized apps and NFTs practical for everyone. The widespread adoption of Layer 2 solutions has reshaped the Blockchain ecosystem. L2s have fueled a “second wave” of DeFi adoption by offering cheaper, faster transactions, attracting new investors to major protocols like Aave and Uniswap.⁷⁰ They have also enabled massive growth in the NFT and gaming sectors, with platforms like Immutable X providing gas-free minting and easy user onboarding.

Layer 2 solutions evolve from temporary fixes into a core component of Ethereum’s infrastructure, driven by strategic Layer 1 upgrades like proto-danksharding (EIP-4844).⁷¹ Activated in March 2024, EIP-4844 introduced “blob transactions” that drastically cut Layer 2 transaction costs by 10-100x and boosted Layer 2 transaction volume by 224%.⁷² This Layer 1 optimization for Layer 2 data solidifies their role, suggesting a future where Layer 1 secures data while most transactions happen on Layer 2.

4. Methods

This research adopts a quantitative approach and employs an experimental design methodology. It operates the voting systems with synthetic population data using open-source software. The dataset includes citizens’ National IDs and basic demographic information and is publicly accessible through the provided URL [1].

Oman’s electronic voting system is structured around diverse system models and innovative methods to enhance the efficiency, transparency, and accessibility of electoral processes nationwide. These models encompass centralized, decentralized, and hybrid approaches, each tailored to address specific requirements and challenges within Oman’s electoral framework.⁷³ In the centralized model, a single authority oversees all aspects of the electronic voting process, ensuring uniformity and centralized control over voter registration, ballot distribution, and result aggregation.⁷⁴ The proposed model emphasizes centralized security measures and operational oversight to safeguard against potential threats and ensure the integrity of the electoral outcome. Conversely, decentralized systems distribute authority and processing power across multiple nodes or centers within Oman. This approach enhances resilience against localized disruptions or attacks, as voting operations can continue independently across different regions.⁷⁵ Decentralized models, such as Blockchain, often require robust coordination mechanisms to synchronize voter data and ensure consistent standards of security and accuracy throughout the electoral process. Addressing these challenges is crucial for improving elections’ efficiency, security, and inclusivity, with Blockchain technology offering potential solutions to many of these issues.⁷⁶ Hybrid systems in Oman blend elements of both centralized and decentralized models, offering flexibility and redundancy while maintaining centralized oversight over critical electoral functions. This model allows for adaptive responses to varying operational conditions and enhances the overall reliability and resilience of the voting system. Key methods employed in Oman’s electronic voting system include advanced biometric authentication technologies to verify voter identities securely. Biometric methods such as fingerprint scanning or facial recognition help mitigate fraud risks and ensure that each vote is cast by the rightful individual, enhancing overall confidence in the electoral process.

The proposed model based on Blockchain technology plays a pivotal role in Oman’s electronic voting system by providing a secure and transparent method of recording and verifying votes. The system helps to ensure their security and reliability, as existing e-voting systems lack a clear protocol, making it difficult to ensure their security and reliability. The system utilizes the Blockchain’s distributed ledger to securely record and verify each vote. Unlike traditional centralized voting systems, where trust is placed in a single authority, Blockchain-based voting ensures that all transactions (such as voter registration, validation, voting, and counting) are transparently recorded and cannot be altered retroactively. Each vote is cryptographically secured, linked to previous transactions, and distributed across a network of computers (nodes), making it extremely difficult for malicious actors to manipulate voting data. Blockchain’s decentralized ledger ensures the immutability and transparency of voting records, making them highly resistant to tampering or manipulation and eliminating the need for intermediaries and central authorities, fostering trust among participants and reducing the risks of fraud or coercion. This aspect is particularly valuable in participatory voting systems, where ensuring the integrity of each vote is paramount to maintaining the democratic process’s legitimacy. The decentralized storage of voting data ensures the integrity and confidentiality of voting data, and Oman’s electronic voting system employs robust encryption techniques and stringent security protocols. These measures protect sensitive information from unauthorized access or manipulation, maintaining the privacy and trustworthiness of the electoral process. In addition, audit mechanisms allow for the verification and scrutiny of all actions taken during the voting process, ensuring adherence to established protocols and regulations.

¹<https://url.us.m.mimecastprotect.com/s/LJ24CkRwomf5wBgE7T2fyTGMWiA?domain=figshare.com>

Ethereum offers a highly secure and transparent foundation for electoral processes, ensuring trust in digital voting systems. Its smart contract automation eliminates the need for intermediaries, reducing human errors and minimizing the risk of election fraud.³³ By leveraging a decentralized network, Ethereum ensures that no single entity controls the voting process, making tampering or manipulation virtually impossible. Additionally, the immutability of its Blockchain enhances public confidence by recording every vote on a transparent ledger that cannot be altered.

Ethereum has limitations regarding transaction scaling, a factor critical to the effective, secure, and economic infrastructure underpinning voting systems.⁷⁷ An estimated 15-30 transactions per second (TPS) is what Ethereum can achieve succinctly in its current settings, which is abysmally low for elections with millions of voters. This too unrestricted scenario would cause a complete and catastrophic collapse for the network, higher fees and processing delays would cause significant issues, and slower vote acceptance mechanisms would result in security risks.⁷⁸ Ethereum's network has faced catastrophic congestion spikes in high-demand periods. Such an election requires consolidating Ethereum's voting systems, which require instantaneous and secure logging of millions of votes. Crippling slowdowns under such intense pressure could result in a complete loss of voter faith, raising alarm bells about the legitimacy of the automotive voting process. Without vast alterations, the inefficient management of explosive transaction volume would render changing publicized elections without an overhaul infeasible.

Furthermore, Ethereum's ongoing scalability improvements, particularly through Layer 2 solutions such as Rollups and Polygon, significantly enhance its transaction speed and cost efficiency, making it more viable for large-scale elections.⁷⁸ Research and developers have been focusing on Layer 2 scaling solutions and sharding, which seek to optimize transaction load distribution, as potential approaches to alleviating Ethereum's scalability issues. Additional protocols that increase transaction throughput and are based on the main Ethereum Blockchain are called Layer 2 protocols (L2). The optimistic rollups assume all submitted transactions are valid without preliminary checks and only run computational checks if there is a dispute. They reduce expenses like ZK Rollups perform off-chain transaction verifications before final on-chain submissions to ensure that only legitimate submissions are included.⁷⁹ Rollups involve combining several transactions into a single one, and this combined transaction is what is submitted to the main Blockchain. A subset of voters can initiate and process transactions off-chain; only the final result is committed to Ethereum. This provides considerable relief to the network. Blockchains other than Ethereum that run in parallel to Ethereum are called sidechains. They can maintain their ledgers and, in conjunction with, periodically merge with the primary Ethereum Blockchain. This sidesteps congestion.

Parallel processing through shards is yet another potential solution. It divides the Ethereum Blockchain into smaller constituent chains (shards) that can individually execute transactions and manage smart contracts. It permits parallel execution and increases scalability while ensuring that voter transactions are processed seamlessly regardless of network congestion.⁸⁰

The system's use case model further explains its concepts and key actors that make it successful.^{13,53}

4.1 Use case model

The proposed Blockchain-based voting system involves key actors: election administrators, voters, election auditors, Ethereum nodes, and smart contract developers. Election administrators manage the process, voters cast secure electronic votes, auditors ensure integrity, Ethereum nodes maintain the network, and developers create smart contracts. **Figure 4** illustrates a simplified use-case model of the system, depicting interactions among these actors and the Blockchain infrastructure, enhancing transparency and security in electoral processes.

Election Administrators The main operations of election administrators are as follows:

- **Manage Election:** Administrators can create, configure, and manage election campaigns. This includes setting up the parameters such as voting dates, eligible candidates, and voter eligibility criteria.
- **Monitor System:** Administrators can monitor the overall system health, including Blockchain network status, transaction verification, costs, and any potential security threats.
- **Generate Reports:** Administrators may need to generate various reports on election results, voter turnout, and system performance for auditing and analysis.
- **Oversee Election Process:** Election authorities can oversee the entire election process, ensuring fairness, transparency, and compliance with electoral regulations.

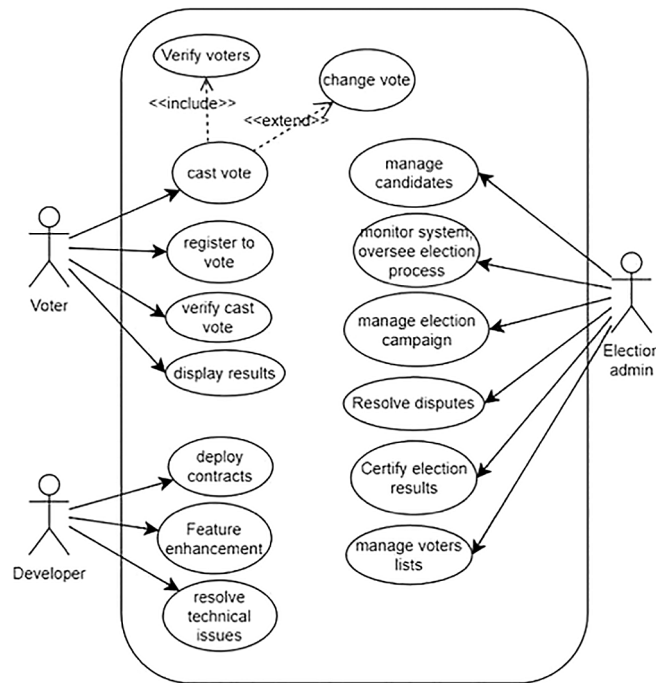


Figure 4. Use case model of the system.

- **Resolve Disputes:** Election authorities can intervene and resolve disputes or challenges related to voter eligibility, ballot integrity, or other election-related issues.
- **Certify Election Results:** Election authorities can certify the final election results based on the data recorded on the Blockchain ledger, providing official validation of the outcome.

Voters Voters are the general public eligible for a voting campaign. The main operations of voters are as follows:

- **Register to Vote:** Voters can register themselves on the Blockchain-based system to participate in elections. This process may involve identity verification and confirmation of eligibility criteria. To simplify the process, voters can register to the Blockchain system for a voting campaign via an intermediate web application server operated by a government body such as the election commission.
- **Cast Vote:** Voters can securely cast their votes for their chosen candidates. The system should ensure anonymity, integrity, and verification of each vote.
- **Verify Vote:** Voters can verify that their vote has been correctly recorded on the Blockchain ledger, providing transparency and confidence in the voting process.
- **View Candidate's details:** Voters can access information about candidates, their agendas, and other relevant details to make informed voting decisions.

Auditors

- **Audit System Integrity:** Auditors can verify the integrity of the smart contracts and voter registration, ensuring that the system operates without any manipulation or fraudulent activities.
- **Review Transaction History:** Auditors can review the transaction history stored on the Blockchain ledger to validate the accuracy and transparency of the voting process.
- **Verify Compliance:** Auditors can ensure the system complies with relevant regulations and standards governing elections, data privacy, and security.

Web Application and Smart-Contract Developers

- **Develop Smart Contracts:** Developers can create and deploy smart contracts that govern the rules and logic of the voting system, ensuring transparency, security, and automation of voting processes.
- **Enhance System Features:** Developers can propose and implement enhancements to the Blockchain-based voting system to improve scalability, usability, and security.
- **Resolve Technical Issues:** Developers can troubleshoot and resolve technical issues within the system, such as bugs, performance bottlenecks, or vulnerabilities.

Ethereum Nodes Ethereum nodes play a prominent role in the Blockchain-based voting system by actively listening for and processing critical transactions such as voter verification, vote casting, and election result dissemination. Their active participation is integral to maintaining the integrity and transparency of the electoral process. Therefore, Ethereum nodes are considered principal actors within our system, ensuring secure and decentralized management of voting operations. This collaborative effort among nodes strengthens the reliability and resilience of the voting system, bolstering confidence in the accuracy and fairness of election outcomes.

- **Verify voters:** In Ethereum-based voting systems, nodes verify voter identities before they can cast their votes. Nodes maintain the Blockchain's integrity by executing smart contracts that authenticate voters and record their votes securely. This decentralized approach ensures transparency and safeguards against manipulation, promoting trust in the electoral process.
- **Receive cast votes:** In Ethereum-based voting systems, Ethereum nodes receive and validate votes cast by voters for candidates, ensuring the integrity and transparency of the voting process through decentralized verification and Blockchain recording.
- **Display/Print results:** In Ethereum-based voting systems, Ethereum nodes validate votes, tally election results, and securely record the outcome on the Blockchain, ensuring transparency and immutability in the electoral process.

System Architecture The architecture of a simplified Blockchain-based distributed voting system comprises four tiers: the storage services tier (Tier-1) ensures secure storage of encrypted voter data and transaction records, collectively enhancing transparency, security, and reliability in electoral operations. The Web3 Blockchain network smart contract services tier (Tier-2) hosts smart contracts governing the voting process. The Web2 application services tier (Tier-3) manages business logic and user authentication. The front-end tier (Tier-4) provides user interfaces for voter interactions.

Tier-1

As illustrated in [Figure 5](#), tier-1 encompasses critical data storage services like MySQL for traditional data management and Blockchain ledger for immutable, decentralized record-keeping. Tier-3 voting application server utilizes MySQL-based storage engines to efficiently store and retrieve user credentials, voter information, and administrative data. Meanwhile, the Ethereum network in tier-2 involves Ethereum nodes maintaining individual copies of the Blockchain ledger that guarantees transparency, security, and integrity in the voting process by securely recording each vote as a tamper-proof transaction. This Tier-1 layer forms the foundation of the application's architecture, combining the reliability of traditional databases with the innovation and trust of Blockchain technology to uphold the integrity of the voting process. In other words, a Blockchain ledger ensures that a voter can only cast a single vote for a candidate by restricting the backtracking of the cast votes to the candidates and producing trustworthy voting campaign results.

Tier-2

Tier 2 is the critical service layer of a trust-less distributed voting system. The trust-less means the system is designed so that participants don't have to trust a single, central authority to function securely and reliably. It is composed of an Ethereum Blockchain network. An Ethereum network consists of thousands of computing nodes that individually store a copy of a distributed, synchronized database known as the Blockchain. The network is a distributed, peer-to-peer network.⁸¹ The major feature of such a network is that each Ethereum node can store smart contracts in the Blockchain and run them in its Ethereum virtual machine. A smart contract is a stored procedure that performs atomic digital

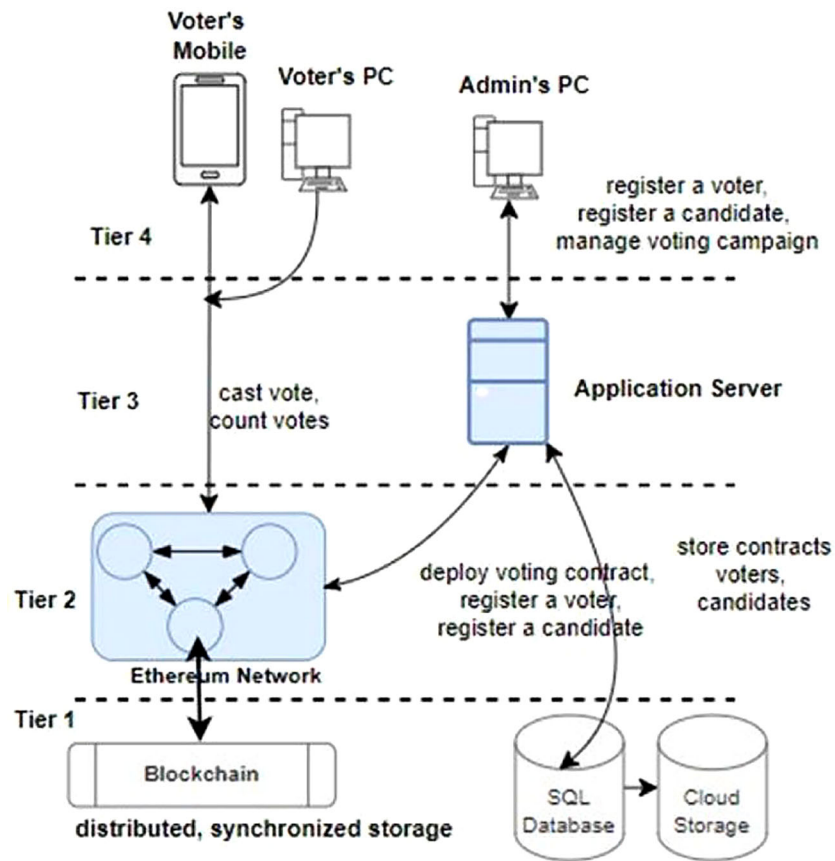


Figure 5. A simplified Blockchain-based distributed voting system architecture.

transactions, the outcome of which can be verified by the participating Ethereum nodes and committed to the Blockchain database. Every transaction is stored in the Blockchain network, such as Ethereum, and such storage is immutable and tamper-proof.⁸²

The Ethereum network provides the following services to our voting system:

1. Register voters and store them in the Blockchain.
2. Register and store election officers and election administrator mandates in the Blockchain.
3. Register a valid vote cast by a registered voter for a candidate
4. Respond to the user's query to check if a voter is registered.
5. Respond to the user's query to check the count of votes received by a candidate.

The operations are performed in the smart contract deployed on the Ethereum network.

Tier-3

Tier 3 is composed of the web application services that can be utilized by election administrators for the following tasks:

1. Register the voting campaign officers.
2. Register the voters for the voting.
3. Register the candidates for the voting campaign.
4. Deploy the voting campaign smart contracts to the Ethereum Blockchain.

Tier-3 application servers utilize Tier-1 relational database services such as MySQL for data storage. Our tier-3 web application service was built in C#. Alternatively, we can use cloud storage services such as database.com, Oracle, Amazon Aurora, IBM Db2, and BigQuery.⁸³

Tier-4

Tier 4 architecture encompasses various end-user devices, including smartphones, personal computers, tablets, and client applications such as web browsers and crypto-wallets. These devices play a pivotal role in the Blockchain-based voting system by serving as interfaces through which stakeholders, including voters and election administrators, interact with the electoral process. One of the distinctive features of devices in this layer is their ability to securely host and manage crypto wallets. Crypto wallets store cryptographic keys that authenticate users and authorize their actions within the voting system. For voters, these wallets enable secure and anonymous casting of votes, ensuring confidentiality and integrity throughout the voting process.

Election administrators utilize crypto wallets to manage voter registration, monitor election activities, and validate results, maintaining transparency and accountability in electoral operations. Moreover, end-user devices in tier 4 facilitate seamless connectivity to the Web3 Blockchain network, where smart contracts execute and enforce the voting process rules. This direct interaction with the Blockchain ensures that all transactions, from voter registrations to ballot submissions and result tabulations, are recorded immutably and transparently on the distributed ledger. By leveraging the capabilities of smartphones, computers, and other client applications, the voting system enhances accessibility for voters, enabling them to participate conveniently from any location with internet access. This inclusivity fosters higher voter turnout and engagement in democratic processes, while the secure handling of crypto wallets and Blockchain interactions ensures the system's resilience against cyber threats and fraud. The voters utilize the client applications for the following tasks:

1. To cast the votes for the candidates.
2. To register for the voting campaign.

We created programs based on .Net Core 7 to simulate the operations of the tier-4 devices and applications. However, browser-based client applications can be developed for production purposes to support multiple operating systems, such as Windows, MacOS, Linux, Android and iOS.

5. Experimentation setup

Researchers developed a simulation package in the C# .Net platform to determine the proposed system's efficiency. The smart contract was developed in C# and deployed in the Ethereum-based Geth test chain provided by Netherium. The Blockchain network Geth is deployed on a machine with Microsoft Windows 10 Enterprise OS, a 12th Gen Intel® Core™ Intel i7-12700, 2100 Mhz, 12 Cores processor, and 16GB installed Physical Memory (RAM).

For experimentation, a testbed based was established using the Geth test chain and .Net Core clients, integrated with Ethereum through Netherium a .Net library designed for Blockchain connectivity [2]. The library contains APIs simplifying access and smart contract integration with Ethereum nodes. In the context of communication design, the protocol is HTTP, which can be switched to HTTPS for a secure channel. However, to simplify communication, the protocol was designed to be HTTP-based. The algorithms described in this section were implemented using C#.Net.

⁸³<https://docs.netherium.com/en/latest/>

Algorithm 1. SmartContract π .

1: **Struct:** Election {uint eid; uint year; string name; uint type}

2: **Storage:**

3: address ContractOwner

4: Elections $\leftarrow$ MAP (address $\Rightarrow$ Election);

5: Voters $\leftarrow$ MAP (address $\Rightarrow$ uint);

6: Candidates $\leftarrow$ MAP (address $\Rightarrow$ uint);

7: Voters $\leftarrow$ MAP (address $\Rightarrow$ uint);

8: **procedure** CandidateRegistration (address ID)

9: **if** Candidates [ID] **then**

10: return FALSE

11: **end if**

12: **if** ContractOwner == Message. Sender **then**

13: return FALSE

14: **end if**

15: Candidates [ID] = TRUE

16: **end-procedure**

17: **procedure** VotersRegistration (address ID)

18: **if** Voters [ID] **then**

19: return FALSE

20: **end if**

21: **if** ContractOwner == Message. Sender **then**

22: return FALSE

23: **end if**

24: Voters [ID] = TRUE

25: **end-procedure**

26: **procedure** CastVote (address ID)

27: **if** Voters [ID] == TRUE **then**

28: return FALSE/*voter already voted*/

29: **end if**

30: **if** Candidates [ID] **then**

31: return FALSE/*candidate is registered.*/

32: **end if**

33: Candidates [ID] += 1

34: return TRUE

35: **end-procedure**

36: **procedure** VoteCount (address ID)

37: **if** Candidates [ID] **then**

38: return Candidates [ID]

39: **end if**

40: return FALSE

41: **end-procedure**

42: **procedure** Constructor()

43: ContractOwner = Message.Sender

44: return TRUE

45: **end-procedure**

Algorithm 2. DeploySmartContract.

```

1: Inputs: EthereumNetwork  $\chi \leftarrow (\text{networkid} = \{\}, \text{url} = \{\})$ , SmartContract  $\pi$  /*Hereafter,  $(..)$  is a tuple defining the object*/
    /* $\{..\}$  is a place-holder for appropriate values.*/
2: Outputs: SContractReceipt  $r \leftarrow (\text{time}, \text{contract-address}, \text{message})$ 
3:  $v.\text{connect\_to}(\chi)$ ; /*connect the the blockchain network network  $\chi$ */
4:  $r \leftarrow \chi.\text{deploy}(\pi)$ ; /*execute a transaction to deploy the smart contract  $\pi$  on the blockchain network  $\chi$ */

```

Algorithm 3. CandidateRegistration.

```

1: Inputs: EthereumNetwork  $\chi$ , SmartContract  $\pi$ 
2: Outputs: CRegistrationReceipt  $r \leftarrow (\text{time}, \text{address}, \text{message})$ 
3: Initialize candidates  $C \leftarrow \{c_0, c_1, ..c_n\}$ ;  $c_i \leftarrow (ID, \text{name}, \text{SSN}, \text{dob}, \text{election-id})$ 
4: for  $c$  in  $C$  do:
5:    $c.\text{connect\_to}(\chi, \pi)$ ;
6:    $r \leftarrow c.\text{register}()$ ; /*execute a transaction to register a candidate for an election with ID,  $\text{eid}$ .*/
7: end for

```

Algorithm 4. VotersRegistration.

```

1: Inputs: EthereumNetwork  $\chi$ , SmartContract  $\pi$ 
2: Outputs: RegistrationReceipt  $r \leftarrow (\text{time}, \text{address}, \text{message})$ 
3: Initialize voters  $V \leftarrow \{v_0, v_1, ..v_n\}$ ;  $v_i \leftarrow (ID, \text{name}, \text{SSN}, \text{dob}, \text{election-id})$ 
4: for  $v$  in  $V$  do:
5:    $v.\text{connect\_to}(\chi, \pi)$ ; /*establish the connection*/
    /* to the network and the smart contract*/
6:    $r \leftarrow v.\text{register}()$ ; /*execute a transaction to register a voter for an election with ID,  $\text{eid}$ .*/
7: end for

```

Algorithm 5. CastVote.

```

1: Inputs: EthereumNetwork  $\chi$ , SmartContract  $\pi$ , Voters  $V$ , Candidates  $C$ 
2: Outputs: VoteCastReceipt  $r \leftarrow (\text{time}, \text{address}, \text{message})$ 
3: for  $v$  in  $V$  do:
4:    $v.\text{connect\_to}(\chi, \pi)$ ;
5:    $r \leftarrow v.\text{cast\_voteto}(c: c \in C)$ ; /*voter  $v$  voting the candidate  $c$ */
6: end for

```

6. Results and Discussion**6.1 Comparison of the proposed model with existing solutions**

The model's primary strength lies in its pragmatic and context-specific design, differentiating it from other nations' theoretical frameworks and legacy systems. While we acknowledge the pioneering work in Estonia and the advanced research in South Korea, the proposed model's core innovation is its tailored approach for Oman. It moves beyond a purely academic exercise by seamlessly integrating with Oman's existing e-governance infrastructure, specifically the national ID card and the 'Sawtak' system. This strategic decision minimizes adoption barriers and leverages an already-trusted biometric authentication layer. The focus on a simple, modular design using both synchronous and asynchronous transactions is a deliberate architectural choice that directly addresses Oman's unique logistical challenges, ensuring a system that is not only technologically robust but also practically implementable and regionally relevant.

Estonia's well-established "i-Voting" system, a centralized remote voting solution, has been used for years, authenticating voters through a national ID card with a chip and a Public Key Infrastructure (PKI).⁸⁴ However, despite its success,

the system has drawn criticism from security experts due to its centralized server architecture, which presents a single point of failure and potential vulnerabilities.⁸⁵ The proposed model leverages a decentralized Blockchain, providing higher transparency and auditability. Unlike Estonia's system, where a central authority manages the vote record, our model uses a public ledger that permanently and transparently records votes, allowing anyone to verify the entire voting record.⁸⁶ This immutability builds public trust and offers a more resilient solution by eliminating the single point of failure.

South Korean research, notably from institutions like ETRI, has explored complex, theoretically advanced Blockchain e-voting systems using hybrid consensus models and sharding to achieve high scalability.⁸⁷ In contrast, our model's pragmatic design and immediate implementability are strengths. Instead of relying on novel cryptographic techniques or architectural overhauls, the proposed solution seamlessly integrates with established biometric technology and a robust existing Blockchain (Ethereum). This approach is a practical engineering choice, utilizing a modular, synchronous, and asynchronous model to address real-world logistical challenges and minimize adoption barriers.

The "Sawtak" is an operational system that verifies voter identity off-chain using the National ID and biometrics; its centralized architecture carries inherent risks. The Blockchain-based model addresses these limitations by introducing a decentralized system with voting data distributed across a network of nodes, which eliminates a single point of failure and provides resilience against cyberattacks. The voter's real-world identity is decoupled from the on-chain record; while the public can verify that a valid vote was cast, they cannot determine who cast it. Additionally, the model can scale and use Layer 2 solutions to accommodate a national-scale election.

6.2 Model uniquely suited to the Omani context

The proposed model aims to enhance Oman's existing e-governance infrastructure to integrate with the National ID card and "Sawtak" system's trusted biometric verification. The proposed system helps to shift from the traditional trust us model to a verify for yourself solution. The tamper-proof nature of Blockchain lets anyone audit the entire voting record independently and build public confidence. The proposed model is built with a practical, modular design that tackles a major logistical challenge by handling different jobs at different speeds. For less time-sensitive tasks like voter registration, it works like sending an email; the system doesn't have to wait for a reply to move on. But it switches to a real-time, instant confirmation method for the high-stakes moment of casting a vote on election day. This innovative, two-speed approach keeps the system running smoothly and efficiently without overcomplicating a critical process.

6.3 The scalability challenge of Layer 1 (L1)

The simulation of the proposed model operates on a Layer 1 protocol. It takes advantage of the core functional and security benefits of a Blockchain-based voting system, relying solely on the Ethereum mainnet, which would introduce a significant scalability bottleneck for a national-scale election in Oman. These limitations of Layer 1, such as the number of transactions per second (TPS), could lead to high network congestion, delayed vote processing, and high transaction fees. The issue might trigger more during the peak voting hours, undermining the system's efficiency and public accessibility.

6.4 Integration with Layer 2 (L2) rollups

To overcome this challenge, the proposed model is designed with a future-proof architecture that integrates with Layer 2 scaling solutions, specifically Optimistic and Zero-Knowledge (ZK) Rollups. These L2s are designed to process most transactions off the main chain, bundling them into a single, compressed transaction that is then settled on Layer 1. Integrating with Layer 2 increases transaction throughput while maintaining the core security with assurances of the Ethereum mainnet, which is helpful for the national election in Oman, to handle the scalability. Layer 2 can accommodate the authentication of voters and ballot casting on separate networks for instant transactions with nominal fees. Furthermore, Layer 1 can be used as a decentralized settlement layer to receive cryptographic proofs of all the votes cast. The dual architecture can handle large numbers of voters during national elections with security, transparency, and integrity.

6.5 Zero-Knowledge Proofs (ZKPs)

The digital voting on Public Blockchain offers transparency and auditability, but presents a critical challenge to voter privacy. The advanced analysis of the data in the public ledger can lead to voters' unique wallet addresses associated with their vote. Hence, there is a chance of compromising confidentiality due to the inherent immutability of the public ledger. This tension between transparency and privacy is crucial for consolidating the model's design using Zero-Knowledge Proofs (ZKPs).⁸⁸ The cryptographic method ZKP allows one party to prove to another party that a given statement is true, without revealing any evidence outside the statement's validity.⁸⁹

In the electoral system, voter can cryptographically prove their eligibility and the validity of their vote by hiding their identity or the chosen candidate. It helps to handle privacy concerns of public Blockchains and achieve privacy-preserving decentralized applications. ZKPs can be integrated into the proposed model, and instead of the voter's wallet directly submitting a transaction with their vote to the Blockchain, it will operate in a privacy-centric manner.⁹⁰ The voter can still authenticate using national ID and biometric data via the 'Sawtak' system, establishing eligibility.

After the authentication system generates a cryptographic ZKP to confirm the validity of the voter data and vote regarding the eligibility criteria, the secured data is submitted to the smart contract on the Blockchain. The smart contract would verify the proof's validity without seeing the voter's identity or the specific vote, which would be recorded in an encrypted or anonymized form. The process will help maintain the immutable public record of all the votes cast during the election, ensure transparency, and make the final tally easier. Further, it will guarantee that no external party can link a specific vote back to a particular individual. ZKPs can ensure that the voting model is transparent and privacy-preserving, addressing the fundamental weakness of all public Blockchains, strengthening the system's integrity, and building public trust.

6.6 Performance evaluation

Performance evaluation in a Blockchain-based voting system involves assessing its efficiency, reliability, and scalability, focusing on metrics such as transaction throughput, latency, scalability under increasing loads, security, fault tolerance, resource utilization, audit ability, transparency, and user experience to ensure optimal system operation and voter trust. The research on a Blockchain-based voting system for Oman yielded several key findings from its experimental simulation. The proposed model demonstrated strong technical performance, successfully processing a high volume of simulated voter data and executing both voter authentication and ballot casting with minimal delay via smart contracts. The simulation also confirmed the system's robust security features, showing that its decentralized nature prevented single points of failure and that votes recorded on the Blockchain were immutable and could not be altered or deleted by external parties. Furthermore, the biometric authentication step effectively blocked duplicate votes from a single user.

While these technical findings are conclusive, the paper presents several speculative statements about the system's real-world implementation. These are not proven facts but predictions and hypotheses about what could happen if the system were deployed. For instance, the Blockchain's inherent transparency could lead to greater public trust, increased voter turnout, and higher public confidence in the electoral process. However, these are logical, well-reasoned predictions and not conclusions supported by experimental data, as a real-world social experiment would be required to prove them. Similarly, the study assumes that Oman's nationwide telecommunications infrastructure can support the system, but it cannot definitively confirm this without a separate, large-scale assessment of the real-world network's readiness.

As part of the prototyping, the following tasks were completed:

1. Load a 1000 list of voter-related data to SQL servers. Each voter is identified by a synthetic Social Security Number (SSN) and has other information such as full name, address, phone number, and email address.
2. A client program developed in C# .Net produced 1000 voting (casting votes) transactions encompassing 1000 total voters identified with synthetic Social Security Numbers (SSN) executed against the network.

Two main transactions, voter registration and vote casting, were executed in two different ways:

1. Synchronous transactions: This method synchronously submits transactions to the Blockchain network. In other words, individual voter registration and individual vote-casting transactions were executed one after another. A total of 1000 transactions for each were executed to measure the execution time. Figure 6 shows the time required for registration and voting 20 individual transactions run synchronously.
2. Asynchronous transactions: This method submits transactions to the Blockchain network asynchronously. The voter registration and vote-casting transactions were executed asynchronously in batch sizes of 10, 20, 30, and 50. For example, with a batch size of 10, 10 voter registration transactions and 10 vote-casting transactions were asynchronous. The final vote registration and vote casting time for each batch were measured. Figure 6 shows the time required to execute 1000 transactions for registration and voting. The red plot is the line plot for the time (MS) taken by transaction for voter registration, the grey line plot is for the time (in MS) taken by transaction for voting and receipt of the confirmation, and the blue line plot is the total time taken for both transactions. The spikes in time measurements in asynchronous transactions are often a result of the dynamic and distributed nature of asynchronous systems, where various factors such as network latency, resource contention, flow control queuing delays, and resource allocation and scheduling can influence transaction processing time.

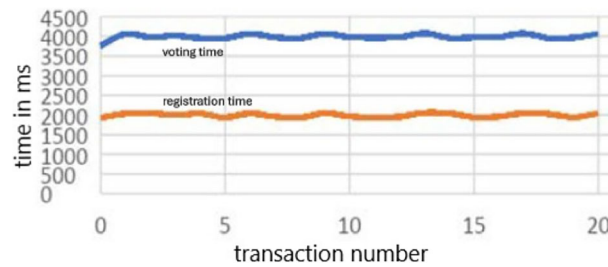


Figure 6. Transactions Times in Millisecond for registration and voting transaction.

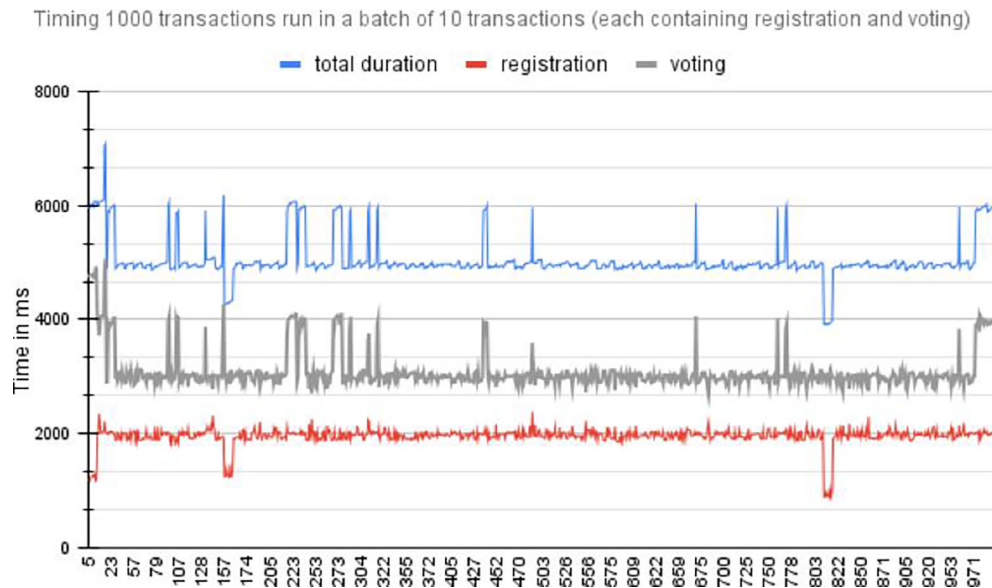


Figure 7. Performance per 1000 Transactions.

Figure 7 shows the time required by asynchronous execution of 100 batches, each consisting of 10 transactions for voter registration and vote casting. The red plot is the line plot for time (MS) taken by a transaction for vote casting and receipt of the confirmation, and the blue line plot is the total time taken for voter registration.

Figure 8 Timing 100 batches of transactions, each containing 10 individual transactions for registration and voting.

Figure 9 displays the average time required for the asynchronous execution of 100 batches, each containing 10 transactions for voter registration and vote casting. This metric is crucial for evaluating the efficiency of the Blockchain-based voting system, particularly in handling concurrent transactions during electoral peaks. It underscores the responsiveness and reliability of end-user devices like smartphones and computers securely processing these transactions, highlighting the system's scalability and performance under varying loads. These insights are essential for optimizing the voting system's infrastructure to ensure robustness and effectiveness in electoral operations.

The experiment primarily focuses on the viability and processing time of election-related transactions. Security-related issues include security vulnerabilities, potential attack vectors, or the risk of Blockchain manipulation. While Blockchain is widely considered a secure and tamper-resistant technology, it is susceptible to several potential vulnerabilities and attack vectors such as Sybil attacks, 51% attacks, Smart Contract Vulnerabilities, data integrity and oracle manipulation, Denial of Service (DoS) attacks, Key Management and Insider Threats, Blockchain forking and inconsistency,⁹¹ among others.⁹² The most pertinent threats are discussed below:

- Sybil Attacks: In a Sybil attack, an adversary creates multiple fake identities to gain disproportionate influence over the network. This can compromise consensus decisions, particularly in permissionless Blockchains. It can be mitigated by enforcing large stakes for validators in a network like Ethereum.⁹³

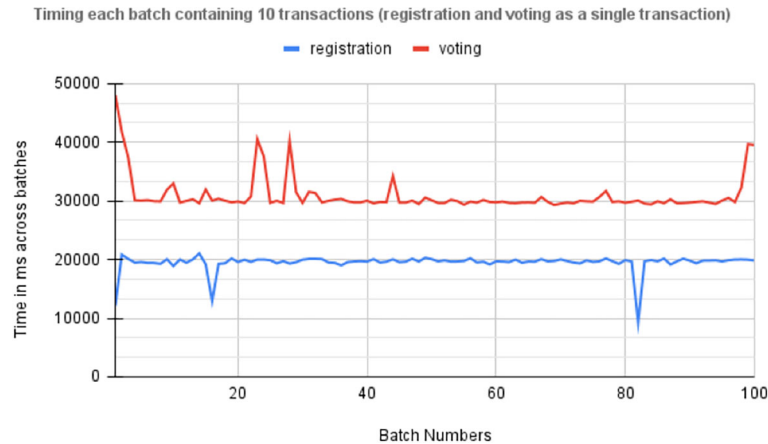


Figure 8. Times across batches each containing 100 transactions.

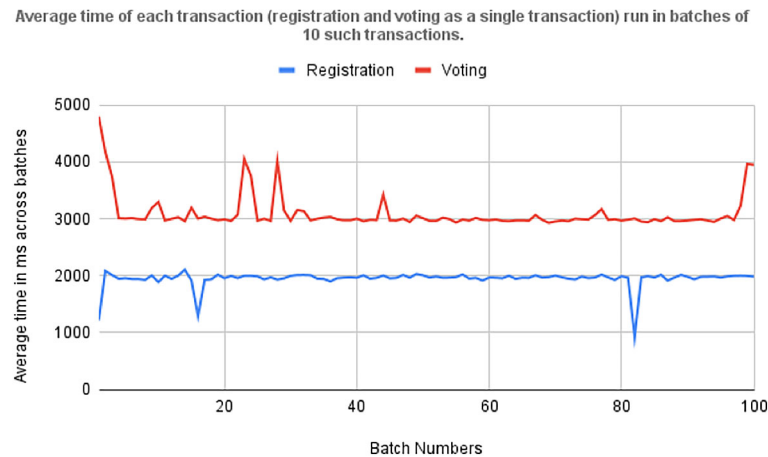


Figure 9. Average time required by asynchronous execution of 100 batches.

- **51% Attacks:** A 51% attack occurs when a single entity or coalition controls most of the network's computational or stake power, allowing them to manipulate the ledger (e.g., double-spending). While this risk is more common in public proof-of-work systems, we can minimize it using a practical Byzantine Fault Tolerant (PBFT) consensus algorithm among pre-approved validators, making it resilient against such attacks.^{91,94}
- **Smart Contract Vulnerabilities:** If smart contracts are used in the framework, they can introduce exploitable bugs (e.g., reentrancy, overflow/underflow errors, or improper access control). We can address this by enforcing rigorous contract auditing and adopting standardized development practices, such as those recommended by OpenZeppelin, and formal verification tools when feasible.⁹⁵
- **Denial of Service (DoS) Attacks:** DoS attacks can target validator nodes or clog the network with excessive transactions. To address this, we can implement rate-limiting, transaction prioritization, and resource-monitoring techniques, especially at the application layer.⁹⁶

7. Conclusion

It has been concluded that Blockchain technology has enhanced the integrity and security of digital voting systems in Oman. Blockchain can significantly bolster voter trust and participation in the electoral process by addressing critical issues such as data integrity, fraud prevention, and transparency. While the advantages of such technology are widely discussed, there is scant evidence detailing public perception, especially among Omani citizens. It is important to know how Omanis view a Blockchain-based voting system because public trust is critical to the success of any election reform. Studies conducted in Oman are lacking, and future work should focus on designing effective questionnaires and trust

measurement scales that accurately determine voters' trust in the technology. Estonia, where its e-governance systems, including its i-Voting system, were bolstered by Blockchain technology. The adoption of Blockchain has improved security and transparency, increasing public trust in digital voting. The Estonia example demonstrates the necessity of solid digital frameworks alongside a digitally connected and trusting populace. The Blockchain-based voting framework ensures a tamper-proof and transparent voting system, maintains voter privacy and facilitates real-time auditing. Successful implementation, however, requires overcoming several challenges, including technological readiness, scalability, and societal resistance to change. The study highlights the necessity for a robust regulatory framework and comprehensive public awareness initiatives to foster acceptance and understanding of Blockchain technology in the electoral context. Policymakers are urged to initiate pilot projects to evaluate the practical implications of Blockchain integration, ensuring a gradual and informed transition. Ultimately, adapting Blockchain in Oman's electoral system could serve as a model for other nations seeking to enhance the security and credibility of their voting processes. Oman has the opportunity to set a precedent in electoral innovation, ensuring that its democratic processes are robust, transparent, and trustworthy.

8. Future work

The proposed model offers numerous benefits, but there are risks and limitations to consider, including the need for advanced infrastructure and technical expertise, potential scalability issues as voter numbers grow, and resistance from the public and stakeholders unfamiliar with or concerned about the reliability of Blockchain technology. Oman must analyze and draw lessons from other global practices to advance to a leading position in Blockchain voting innovation. This goes beyond mobilizing the technology and logistics to examine the socio-political landscape of public trust. By conducting basic perception research on citizens, Oman can customize models compared to existing ones to fit its culture and politics. Such an approach would ensure that all technological changes important in elections would work and be endorsed by the people. The proposed model also has limitations that must be considered [3]. For instance, the issue of scalability needs to be addressed as the number of voters increases substantially. Theoretically, the Ethereum Network supports 119.1 transactions per second (tx/s), which is relatively lower than other Blockchain networks, such as Solana (65000 tx/s), Tron (2516 tx/s), Stellar (1137 tx/s), BNB Chain (2222 tx/s), and Polygon (714.1 tx/s). There are various scaling solutions, including sharding, off-chain computation, rollups (such as Optimistic or ZK-Rollups), and hardware acceleration, among others. Future work includes testing scalability and performance across various Blockchain networks.

First, to address the issue of scalability, our next phase will involve a comprehensive analysis and simulated testing of Layer 2 solutions, such as Optimistic and Zero-Knowledge Rollups. While our current research has successfully validated the core functionality and security on a Layer 1 protocol, a national-scale election requires significantly higher transaction throughput. We will evaluate how L2s can offload the bulk of voter transactions from the main chain, thereby mitigating the risk of network congestion and high fees, while leveraging the Ethereum mainnet's fundamental security.

Second, we will conduct a more in-depth security analysis through formal threat modeling. We plan to incorporate an adversary-centric framework, such as the MITRE ATT&CK for Blockchain, to identify and address a wider range of potential vulnerabilities systematically. This expanded analysis will go beyond traditional Blockchain threats, including collusion among validators, smart contract vulnerabilities, and denial-of-service attacks, ensuring our model is resilient against sophisticated, real-world threats.

Finally, we will design and execute a pilot study with real-world data and comprehensive adversarial stress-testing to bridge the gap between simulation and practice. This pilot will move beyond the limitations of synthetic data to validate the model's performance under live conditions, including fluctuating network traffic and targeted attacks. This crucial step will provide empirical evidence of the system's security and feasibility, serving as a foundation for a full-scale implementation and building public and regulatory trust in a Blockchain-based voting system for Oman.

Ethics and consent

This study does not involve human participants or animals, and therefore, ethics approval was not required. No sensitive data were used, and no real personal information was collected.

Author contributions statement

The authors hereby confirm that we all have made a substantial contribution. A.K.S. has contributed to idea generation and provided direction for the research, and designed the methodology, and wrote the original draft. However, N. A, A.S.

³<https://chainspect.app/dashboard>

S are engaged in the experiment and results. A. N, S. A and H. A have contributed to the partial write-up, visualized the results. All authors reviewed and approved the final version of the manuscript.

Data availability statement

Figshare: Blockchain based secure voting system, <https://doi.org/10.6084/m9.figshare.28113140.v1>.⁹⁷

This project contains the following underlying data:

- Polling folder
- Readme
- Data file – Citizen data_with_national_id.csv

The data supporting the findings of this study are available on under the license [GPL 3.0+](#). All other relevant data are included in the manuscript.

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Open Peer Review

Current Peer Review Status: ? ✓ ✗ ✓

Version 2

Reviewer Report 30 August 2025

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Abdullah Ayub Khan 

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The authors present a technical research paper with relevant topic, proper research methodology and potentially good contribution to the field of studies.

The authors are encouraged to resubmit the paper with more clarity on presented performance assessment metrics with the selected relevant Case studies and possible application scenario with assessment metrics. The paper should be written in proper format, figures should fit within the text, use of font should be uniform in all paper, as well as references should be updated with most recent results. Suggestion and Recommendation:

1. Authors may elaborate more on the novelty/contribution of their work and how it
 2. Authors need to be specific about their problem statement and the scope of their research.
 3. Abstract: elaborate more on the problem statement, findings, and contributions.
 4. Introduction is not clear. Authors may contribute more towards this.
- Contributes to the literature in the second last paragraph of the introduction clearly.
5. Thorough proofreading is recommended.
 6. A few of the figures are taken from the sources and are not cited properly (such as AI Generated), either they may be cited properly with permissions or may be removed/redrawn.
 7. The conclusion is not clear and needs revision and clarity and alignment with the abstract and title."

References:

1. Your references are not listed in good style, as citation style is different from one paper to other.
2. Some of your references are not complete please check.
3. Some citations (references) created in wrong manner (Please follow journal's criteria).

Authors are encouraged to base on recent references about the current development in blockchain technology. Moreover, technology collaborates with

other technologies to create new paradigms, such as artificial intelligence, such machine learning, deep learning, with federated learning.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Yes

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Blockchain

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Reviewer Report 25 July 2025

<https://doi.org/10.5256/f1000research.180515.r400248>

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Farhan Rafat Khan

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Peer Review Comments

1. Novelty and Contribution to Knowledge

The paper presents a blockchain-based model for enhancing electoral integrity in Oman's digital voting system. While blockchain for e-voting is not a novel concept, the application to Oman's electoral context provides regional relevance. The authors contribute by:

- Proposing a **simplified blockchain model** tailored for Oman's voting system.

- Integrating **biometric authentication** with blockchain to strengthen voter verification.
- Evaluating **asynchronous vs. synchronous transactions** for efficiency.

However, the novelty is somewhat limited, as similar frameworks have been proposed for other countries (e.g., Estonia, South Korea). The paper would benefit from a clearer differentiation from prior work, particularly in addressing **Oman-specific challenges** (e.g., public trust, regulatory readiness).

2. Technical Contents

The technical methodology is well-structured, with:

- A **quantitative experimental approach** using synthetic data.
- **Smart contract implementation** (Ethereum-based) for voter authentication and vote recording.
- **Performance evaluation** on transaction times (registration, voting).

Concerns:

- **Scalability:** The paper acknowledges Ethereum's limitations (~15-30 TPS) but does not sufficiently test alternative Layer-2 solutions (e.g., Rollups, Sharding) for large-scale elections.
- **Security Analysis:** While Sybil and 51% attacks are mentioned, deeper threat modeling (e.g., collusion resistance, side-channel attacks) is needed.
- **Real-world Feasibility:** The synthetic data simulation lacks real-world validation (e.g., stress-testing under adversarial conditions).

3. Clarity of Work/Presentation

The paper is **well-organized**, with clear sections and logical flow. However:

- **Figures (e.g., 6-9):** The transaction time plots are useful but could be better annotated (e.g., error margins, statistical significance).
- **Algorithm Descriptions:** The pseudocode (Algorithms 1-5) is clear but lacks complexity analysis (e.g., gas costs for smart contracts).
- **Jargon:** Some blockchain terms (e.g., "trust-less distributed voting") could be simplified for broader readability.

4. Referencing

The references are **relevant and up-to-date**, covering blockchain, e-voting, and Omani electoral processes. However:

- **Missing Key Works:** No discussion of **Zero-Knowledge Proofs** (e.g., ZK-SNARKs) for privacy-preserving voting, despite their growing relevance.
- **Over-reliance on Theoretical Papers:** More empirical studies (e.g., Estonia's i-Voting) would strengthen the practical insights.

5. Comments to Authors

Strengths:

- A **practical blockchain model** with biometric integration.
- **Performance metrics** (async vs. sync transactions) provide actionable insights.

Areas for Improvement:

1. **Expand Security Analysis:** Include formal threat modeling (e.g., MITRE ATT&CK for Blockchain).
2. **Test Scalability Solutions:** Evaluate Polygon or Optimistic Rollups for higher TPS.
3. **Address Privacy:** Discuss how voter anonymity is preserved despite blockchain's transparency.

4. **Compare with Existing Systems:** How does this model improve upon Oman's current "Sawtak" system?

6. Comments to Editor

Accept with Major Revisions. The paper has merit but requires:

- **Deeper technical validation** (e.g., live network testing, adversarial simulations).
- **Stronger differentiation** from prior blockchain voting systems.
- **Policy Implications:** More concrete steps for Omani adoption (e.g., pilot deployment challenges).

If revised thoroughly, this could be a valuable contribution to **applied blockchain research in governance**.

7. Final Decision

Conditional Acceptance – pending revisions addressing scalability, security, and real-world applicability.

Is the work clearly and accurately presented and does it cite the current literature?

Partly

Is the study design appropriate and is the work technically sound?

Partly

Are sufficient details of methods and analysis provided to allow replication by others?

Partly

If applicable, is the statistical analysis and its interpretation appropriate?

Partly

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: cryptography, steganography, cybersecurity, information security, i-voting, information theory

I confirm that I have read this submission and believe that I have an appropriate level of expertise to state that I do not consider it to be of an acceptable scientific standard, for reasons outlined above.

Reviewer Report 20 June 2025

<https://doi.org/10.5256/f1000research.180515.r382046>

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Blerim Rexha

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Final Assessment: Accept with minor recommendation

To make the article scientifically robust and publication-ready, the authors could address the following points:

- Expand literature review to include recent (2024-2025) studies on Ethereum scalability solutions and smart contract security
- Perform explicit statistical analyses on transaction data (confidence intervals, standard deviations, significance testing) and clearly report these statistics to provide more rigorous support for the claims.
- Clearly differentiate between experimentally supported conclusions and speculative statements on infrastructure readiness, societal acceptance, and public trust.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Partly

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Cyber security, electronic voting, Blockchain, privacy

I confirm that I have read this submission and believe that I have an appropriate level of

expertise to confirm that it is of an acceptable scientific standard.

Author Response 01 Sep 2025

Abdul Shaikh

Paper Title: "Blockchain-enhanced electoral integrity: a robust model for secure digital voting systems in Oman"

Responses on Reviewer's Comments by Authors

We would like to thank the expert reviewer for the constructive feedback. We have addressed all the comments and feedback as below.

We believe now our revised manuscript has significantly improved

Reviewer 1 Comment #1

To make the article scientifically robust and publication-ready, the authors could address the following points:

Expand literature review to include recent (2024-2025) studies on Ethereum scalability solutions and smart contract security

Author Response:

Thank you for this valuable and constructive feedback.

To strengthen the scientific robustness of the article, we have expanded the literature review by adding eight recent references (2024–2025) relevant to the latest developments in Ethereum scalability and smart contract security along with adding of two dedicated subsections under Section 3.0 'Related Work':

These additions provide a more up-to-date and comprehensive foundation for our study and align the manuscript with the latest state-of-the-art research.

Reference: Section 3.1 and 3.2 Page # 13-14

Reviewer 1 Comment #2

- Perform explicit statistical analyses on transaction data (confidence intervals, standard deviations, significance testing) and clearly report these statistics to provide more rigorous support for the claims.

Author Response:

Due to data availability constraints, the nature of the dataset, limitations (such as cost for using real-world blockchain network for real the transactions (in the data collection process, or the scope of the current study, which is primarily about the blockchain model for online voting, we are unable to perform such analyses at this time. However, we highly appreciate the comments.

Reviewer 1 Comment #3

Clearly differentiate between experimentally supported conclusions and speculative statements on infrastructure readiness, societal acceptance, and public trust.

Author Response:

Thank you for this helpful observation. To address this observation, we have revised the manuscript by adding clarifying content in Section 6.6 (Performance Evaluation). In this section, we now clearly distinguish between experimentally validated findings (e.g., high throughput of simulated voter data, secure and immutable recording of votes, prevention of duplicate voting through biometric verification) and speculative statements (e.g.,

potential for increased public trust, higher voter turnout, and infrastructure readiness). We explicitly state that these societal and infrastructural aspects remain hypotheses requiring real-world testing and large-scale validation, while the technical performance outcomes are firmly supported by our simulation results. This revision ensures a transparent separation between evidence-based conclusions and forward-looking projections.

Reference: Section 6 Page # 25.

- **Is the work clearly and accurately presented and does it cite the current literature?**

Yes

Author Response:

Thank you

- **Is the study design appropriate and is the work technically sound?**

Yes

Author Response:

Thank you

- **Are sufficient details of methods and analysis provided to allow replication by others?**

Yes

Author Response:

Thank you

- **If applicable, is the statistical analysis and its interpretation appropriate?**

Partly

Author Response:

Addressed

- **Are all the source data underlying the results available to ensure full reproducibility?**

Yes

Author Response:

Thank you

- **Are the conclusions drawn adequately supported by the results?**

Yes

Author Response:

Thank you

Competing Interests: No competing interests were disclosed.

Version 1

Reviewer Report 13 March 2025

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Emilien Paulis

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² University of Luxembourg, Esch-sur-Alzette, Luxembourg

³ University of Luxembourg, Esch-sur-Alzette, Luxembourg

This paper presents a blockchain-based model for enhancing electoral integrity in Oman's digital voting system. It provides a clear rationale for using blockchain technology, details the implementation methodology, and presents experimental results evaluating the model's efficiency and security. The topic is highly relevant, given the growing global interest in secure and transparent e-voting solutions. The study contributes to the field by proposing and testing a blockchain-based electoral model, but certain areas require further elaboration and refinement, especially when reading the article from the perspective of a political scientist.

Conceptual and theoretical framework

The paper provides a general discussion of blockchain technology but, to me, it lacks a strong theoretical foundation that connects blockchain with electoral integrity from a governance and democracy perspective. It would benefit from a more explicit discussion of trust in institutions, voter perceptions, and the role of digital transparency in democratic legitimacy.

Methodology and experimental setting

- While the experimental setup is described in detail, it is unclear how representative the synthetic data is compared to actual voter data in Oman.
- The scalability of the proposed system is not fully addressed: what happens if the voter base significantly increases? Will Ethereum nodes efficiently handle large-scale elections?
- The performance evaluation primarily focuses on processing time but lacks a deeper analysis of security vulnerabilities, potential attack vectors, or risks of blockchain manipulation.
- The choice of Ethereum should be justified further: are there alternative blockchain platforms that might offer better scalability, efficiency, or security?
- How is Oman a particular case (or not) regarding the use of a digital voting system? I'd like the authors to be more reflective about the limit of the case and the external validity of their experimentation.

Conclusion/implications

- The conclusion highlights the importance of public trust but does not provide empirical evidence or survey data on how Omani citizens perceive blockchain-based voting. I would at least open this perspective as future research area.
- The claim that Oman could lead global innovation in electoral blockchain technology is ambitious but not fully substantiated—how does this model compare with existing blockchain-based voting systems in other countries?

Language/writing

- Some sections, particularly the introduction and related work, contain grammatical errors and awkward phrasing, which could be improved for better readability.
- The citations could sometimes be better integrated. Some seem to be listed without critical

engagement or discussion in the text.

Is the work clearly and accurately presented and does it cite the current literature?

Partly

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Not applicable

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Political science

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 14 Apr 2025

Abdul Shaikh

Paper Title: "Blockchain-enhanced electoral integrity: a robust model for secure digital voting systems in Oman"

Responses on Reviewer's Comments by Authors

We would like to thank the expert reviewer for the constructive feedback. We have addressed all the comments and feedback as below.

We believe now our revised manuscript has significantly improved

Reviewer's Comment

- This paper presents a blockchain-based model for enhancing electoral integrity in Oman's digital voting system. It provides a clear rationale for using blockchain technology, details the implementation methodology, and presents experimental results evaluating the model's efficiency and security. The topic is highly relevant, given the growing global interest in secure and transparent e-voting solutions. The study contributes to the field by proposing and testing a blockchain-based electoral model, but certain areas require further elaboration and refinement, especially when

reading the article from the perspective of a political scientist.

Author Response

Thank you for the supportive comments.

Reviewer's Comment

Conceptual and theoretical framework

- The paper provides a general discussion of blockchain technology but, to me, it lacks a strong theoretical foundation that connects blockchain with electoral integrity from a governance and democracy perspective. It would benefit from a more explicit discussion of trust in institutions, voter perceptions, and the role of digital transparency in democratic legitimacy.

Author Response

Reference:

\subsection{Blockchain with Electoral Integrity from a Governance and Democracy Perspective}

Blockchain technology has captured their attention as scholars and policymakers seek to bolster integrity in elections. Dominating the conversations was the impact of digital technology on institutional trust and its relation to electoral systems and the confidence voters placed in those systems \cite{b55}. The available literature reveals that implementing blockchain in voting systems has advantages and challenges.

Defenders of blockchain note that its features, particularly decentralization and the inability to alter information, can remarkably enhance the transparency of elections. Ohize et al. \cite{b52} stress that mobile voting systems based on blockchain technology have the potential to eliminate voter fraud and increase turnout by providing a secure and verifiable system for voting. They highlight that blockchain-based voting systems can be audited by anybody, enhancing credibility and faith in the system and electoral framework. Regardless of these advantages, Assouline et al. \cite{b56} argue that some privacy issues, cybersecurity, digital skills, and the exclusion of certain demographics add to the criticism of using blockchain technology in elections. There needs to be sufficient public awareness and development of the relevant technology to enable everyone's participation.

Similarly, Farooq et al. \cite{b53} explain how blockchain can help protect voter registration information by making alterations transparent and traceable. Jafar et al. \cite{b58} discussed that electronic voting machines with blockchain technology could issue a printed paper receipt to the voters, which would aid in audits and boost trust in the vote-counting process. Moreover, Olaniyi et al. \cite{b57} underscore the risks of applying new technologies to the election process. They contend that elections are critical undertakings wherein things cannot go wrong since poorly applied blockchain systems would lose public trust. Implementing any technological application should be preceded by robust public awareness campaigns managed by reputable, non-partisan election officials.

The use of blockchain in the 2023 elections in Guatemala is an example that illustrates the expected results of the implementation. As stated in a Democracy and Society \cite{b54} report, the technology was applied to restore trust in the electoral system after allegations of fraud in preceding elections. Storing each transaction in a ledger using blockchain technology boosted the electoral system by easily detecting attempted manipulations.

One of the distinguishing features of democracy is transparency and digital democracy. As noted by many researchers, blockchain technology requires a guarded approach regarding security and public trust \cite{b59}. These two elements demonstrate the utmost necessity in maintaining the public's trust in the system. Transparency of information coupled with accessibility indicates an open, robust form of government, which can be achieved through Internet technology. With innovative ideas purchasable through the internet, civic engagement engages us to welcome and engage in democracy through active voting. The exercise of voting becomes a civic duty for citizens of democratic countries as it incredibly directly sources their choices and decisions.

In this regard, elections are a striking example of implementing civic engagement. However, they tend to ignore that civic engagement can be defined in a multi-dimensional framework that features systems thinking that captures the complexity of social systems, making a participatory approach more appealing. Regardless of who utilizes the notion of digital democracy alongside open government, the digitally achieved or rather opened democracy acts as a form of power that the government can exercise and aid in refining the understanding of democracy through freedom.

\subsubsection{Theoretical Foundations and Potential Benefits}

A blockchain functions as a transparent public ledger incapable of modification, which opens the possibility for a new method of ensuring fair elections. Employing specific coding techniques, voting systems based on blockchain technology can offer End-to-End Verifiable (E2E) elections \cite{b60}. E2E implies that votes are cast on time, midstream, stored, and counted while stored. This capability can resolve problems associated with election fraud, risk of hacking, and personnel error. However, adding blockchain does not automatically provide more trust in elections \cite{b61}. Rather, the trust in blockchain voting is founded on the perception of voters, political leaders, and election officers. Without trust and knowledge of such systems, citizens cannot endorse democracy.

\subsubsection{Trust in Institutions and Electoral Legitimacy}

Trust in institutions is one of the central issues in political science. To what degree do people consider that political institutions are executing their roles properly? Standard voting systems dependent on institutions like election commissions, political parties, and independent monitors for free and fair elections rely on multiple functions working synergistically to acquire an equilibrium bloc \cite{b62}. It's more a question of voters' trust in the technology than in existing institutions and whether blockchain technology, which offers a mechanism without a need for a central authority, will make freer elections

possible.

Most studies indicate a general decline in confidence in voting systems across the globe. People are apprehensive regarding cheating, vote tampering, and cybersecurity threats \cite{b63,b64,b65}. Blockchain is suggested as an answer to this problem because there is no need to trust a central authority. All the transactions (votes) can be independently verified. Political analysts say elections are primarily social and not purely technical \cite{b66}. If people remain doubtful about blockchain security, ease of use, and the possibility of government tampering, increased reliability of elections will not be guaranteed.

Research indicates mixed results on users' trust in voting via the Internet. Some suggest that younger, more adept technology users will trust elections run on blockchains while older users grapple with the system's clarity and transparency. Moreover, in cases where there is extreme political division within parties, new election platforms are likely to be more divisive than unifying.

\subsubsection{Voter Perceptions and Digital Transparency}

Public trust in the fairness and accessibility of the electoral processes is fundamental to their credibility. A blockchain framework could worsen fears and misunderstandings without clarity, education, and transparency.

\begin{itemize}

\item \textbf{Perceived Opacity vs. Actual Transparency}

Although blockchain can improve transparency by allowing voting processes to be publicly verified, most voters lack the technical skills necessary to audit blockchain systems \cite{b67,b68}. Without straightforward verification mechanisms, blockchain implementation could inadvertently generate a black-box scenario where the community must depend on trusted specialists to navigate the opaque system instead of analyzing the results independently.

\item \textbf{Digital Literacy and Accessibility}

The positive impact of blockchain technology on electoral integrity relies on the assumption that all voters have the requisite digital literacy skills to engage with a blockchain voting system. Nonetheless, some studies indicate that the digital divides informed by education, age, and socioeconomic status can extend existing disenfranchisement \cite{b69}. That said, implementing blockchain voting needs to be preceded by efforts to educate voters about the technology so that they have adequate understanding and trust in the system.

\item \textbf{Role of Social Media and Disinformation}

The capability of blockchain technology to secure electoral data does not address the challenges of misinformation. Today's elections are as much about the social media narratives, spin, and misinformation surrounding them as they are about casting votes

\cite{b70}. Even if there were a voting system based on blockchain that was utterly impenetrable, trust in the system would still be undermined in highly polarised elections due to rampant inaccurate claims regarding the results.

\end{itemize}

\subsubsection{The Role of Digital Transparency in Democratic Legitimacy}

Democratic legitimacy defines whether a government has the right to rule and if it acts with the consent of the citizens. Improving digital transparency with blockchain technology may help in improving legitimacy by:

\begin{itemize}

\item Offering tangible proof that hacking did not occur with the voting.

\item Minimizing dependence on authoritative sides reduces the chance of bias from the institutions.

\item Increasing the possibility of international observation of elections, allowing outsiders to witness the processes live.

\end{itemize}

Elections conducted using blockchain technology may expose previously undetected discrepancies in the voting process. These discrepancies, however, could be misinterpreted as fraudulent activity, thus eroding trust in the electoral system. While blockchain may reinforce election integrity, its effectiveness will hinge on public perception, institutional support, and integration into current electoral systems \cite{b71}. Should insufficient public confidence and trust exist, an inclusive strategy could lead to opposition to the elections. Lastly, to bolster the democratic image of a nation, issues surrounding democratic election integrity need comprehensive public trust initiatives and civic education campaigns that examine governance issues, establish trust in the system, and systematically approach the integration of blockchain technology.

***Reviewer's Comment**

Methodology and experimental setting

- While the experimental setup is described in detail, it is unclear how representative the synthetic data is compared to actual voter data in Oman.

Authors Response

Reference \section{Methods}

The research study generated synthetic data for the experimental purpose using Python and code is shared with additional material. Authors ensured its validity and accuracy, as it undergoes rigorous data integrity checks to ensure consistency and reliability. These checks include range validation, ensuring all values fall within acceptable limits, such as reasonable demographic details; format validation, confirming that data structures like national IDs adhere to predefined formats; and presence checks, ensuring all necessary fields are present without omissions. By leveraging these checks, the synthetic data mirrors key attributes of real-world population demographics while maintaining privacy and security. Such validation ensures the data's suitability for modeling the blockchain-based voting system, allowing the experiment to test functionalities and performance metrics effectively

while safeguarding against data errors and inconsistencies. This robust validation approach makes the synthetic data both dependable and fit for the study's purposes.

***Reviewer's Comment**

- The scalability of the proposed system is not fully addressed: what happens if the voter base significantly increases? Will Ethereum nodes efficiently handle large-scale elections?

Authors Response

We have updated the following lines in the conclusion section to address the above concern
Reference \section{Conclusion}

While the proposed model offers numerous benefits, it also has limitations that need to be considered. For instance, the issue of scalability needs to be addressed as the number of voters increases substantially. Theoretically, the Ethereum Network supports 119.1 transactions per second (tx/s) (footnote 1), [1] which is relatively lower than other blockchain networks, such as Solana (65000 tx/s), Tron (2516 tx/s), Stellar (1137 tx/s), BNB Chain (2222 tx/s), and Polygon (714.1 tx/s). There are various scaling solutions, including sharding, off-chain computation, rollups (such as Optimistic or ZK-Rollups), and hardware acceleration, among others. Future work includes testing scalability and performance across various blockchain networks.

***Reviewer's Comment**

- The performance evaluation primarily focuses on processing time but lacks a deeper analysis of security vulnerabilities, potential attack vectors, or risks of blockchain manipulation.

Authors Response

We have added the following content at the end of the "Result and Discussion" section to address the above concern

The experiment is primarily conducted to focus on the viability and processing time of election-related transactions. There are security-related issues, such as security vulnerabilities, potential attack vectors, or the risk of blockchain manipulation. While blockchain is widely considered as a secure and tamper-resistant technology, it is susceptible to several potential vulnerabilities and attack vectors such as sybil attacks, 51% attacks, Smart Contract Vulnerabilities, data integrity and oracle manipulation, Denial of Service (DoS) attacks, Key Management and Insider Threats, Blockchain forking and inconsistency (Citation-1), among others (<https://owasp.org/www-project-smart-contract-top-10/>). The most pertinent threats are discussed below:

- Sybil Attacks: In a Sybil attack, an adversary creates multiple fake identities to gain disproportionate influence over the network. This can compromise consensus decisions, particularly in permissionless blockchains. It can be mitigated by enforcing large stakes for validators in a network like Ethereum (Citation-2)
- 51% Attacks: A 51% attack occurs when a single entity or coalition controls most of the network's computational or stake power, allowing them to manipulate the ledger (e.g., double-spending). While this risk is more common in public, proof-of-work systems, we can minimize it by using a practical Byzantine Fault Tolerant (PBFT) consensus algorithm among pre-approved validators, making it resilient against such attacks (Citation-1).
- Smart Contract Vulnerabilities: If smart contracts are used in the framework, they can introduce exploitable bugs (e.g., reentrancy, overflow/underflow errors, or improper access control). We can address this by enforcing rigorous contract auditing and

adopting standardized development practices, such as those recommended by OpenZeppelin and formal verification tools when feasible (Footnote for this web link:<https://ethereum.stackexchange.com/questions/83542/verifying-erc20-contract-that-uses-openzeppelins-library-contracts>).

- Denial of Service (DoS) Attacks: DoS attacks can target validator nodes or clog the network with excessive transactions. We can implement rate-limiting, transaction prioritization, and resource-monitoring techniques to address this, especially at the application layer (Citation-3).

The choice of Ethereum should be justified further: are there alternative blockchain platforms that might offer better scalability, efficiency, or security

Authors Response

Reference \section{Methods and Results}

Ethereum offers a highly secure and transparent foundation for electoral processes, ensuring trust in digital voting systems. Its smart contract automation eliminates the need for intermediaries, reducing human errors and minimizing the risk of election fraud \cite{b72}. By leveraging a decentralized network, Ethereum ensures that no single entity controls the voting process, making tampering or manipulation virtually impossible. Additionally, the immutability of its blockchain enhances public confidence by recording every vote on a transparent ledger that cannot be altered. In terms of scalability, Ethereum Network supports 119.1 transactions per second (tx/s) (footnote 1), which is relatively lower than other blockchain networks, However, there are various scaling solutions, including sharding, off-chain computation, rollups (such as Optimistic or ZK-Rollups), and hardware acceleration, among others. Future work can be include testing scalability and performance across various blockchain networks.

- How is Oman a particular case (or not) regarding the use of a digital voting system? I'd like the authors to be more reflective about the limit of the case and the external validity of their experimentation.

*Reviewer's Comment

Conclusion/implications

- The conclusion highlights the importance of public trust but does not provide empirical evidence or survey data on how Omani citizens perceive blockchain-based voting. I would at least open this perspective as future research area.
- The claim that Oman could lead global innovation in electoral blockchain technology is ambitious but not fully substantiated—how does this model compare with existing blockchain-based voting systems in other countries?

Author's response

Thank you for your valuable comments

The conclusion has been revised as below to address the above concern

It has been concluded that blockchain technology has enhanced the integrity and security of digital voting systems in Oman. Blockchain can significantly bolster voter trust and participation in the electoral process by addressing critical issues such as data integrity, fraud prevention, and transparency. While the advantages of such technology are widely discussed, there is scant evidence detailing public perception, especially about Omani citizens. It is important to know how Omanis view a blockchain-based voting system because public trust is critical to the success of any election reform. Studies conducted in Oman are lacking, and future work should focus on designing effective questionnaires and

trust measurement scales that accurately determine voters' trust in the technology. Estonia, where its e-governance systems, including its i-Voting system, were bolstered into blockchain technology. The adoption of blockchain has improved security and transparency, increasing public trust in digital voting. The Estonia example demonstrates the necessity of solid digital frameworks alongside a digitally connected and trusting populace. The blockchain-based voting framework ensures a tamper-proof and transparent voting system, maintains voter privacy, and facilitates real-time auditing. Successful implementation, however, requires overcoming several challenges, including technological readiness, scalability, and societal resistance to change. The study highlights the necessity for a robust regulatory framework and comprehensive public awareness initiatives to foster acceptance and understanding of blockchain technology in the electoral context. Policymakers are urged to initiate pilot projects to evaluate the practical implications of Blockchain integration, ensuring a gradual and informed transition. Ultimately, adopting blockchain in Oman's electoral system could serve as a model for other nations seeking to enhance the security and credibility of their voting processes. Oman has the opportunity to set a precedent in electoral innovation, ensuring that its democratic processes are robust, transparent, and trustworthy. While the proposed model offers numerous benefits, there are risks and limitations to consider, including the need for advanced infrastructure and technical expertise, potential scalability issues as voter numbers grow, and resistance from the public and stakeholders unfamiliar with or concerned about the reliability of Blockchain technology. Oman must analyze and draw lessons from other global practices to advance to a leading position in blockchain voting innovation. This goes beyond mobilizing the technology and logistics to examine the socio-political landscape of public trust. By conducting basic perception research on citizens, Oman can customize models compared to existing ones to fit its culture and politics. Such an approach would ensure that all technological changes important in elections would work and be endorsed by the people.

Language/writing

- Some sections, particularly the introduction and related work, contain grammatical errors and awkward phrasing, which could be improved for better readability.
- The citations could sometimes be better integrated. Some seem to be listed without critical engagement or discussion in the text.

Author's response

Thank you for your valuable comments

We have thoroughly revised the introduction and related work sections to correct grammatical errors and improve the overall clarity and flow of the text. Also, we have carefully reviewed and revised the the references and citations throughout the manuscript. Where appropriate, we have added further discussion and critical engagement to ensure that cited works are meaningfully connected to our contributions.

- **Is the work clearly and accurately presented and does it cite the current literature?**

Partly

- **Is the study design appropriate and is the work technically sound?**

Yes

- **Are sufficient details of methods and analysis provided to allow replication by others?**

Yes

- **If applicable, is the statistical analysis and its interpretation appropriate?**

Not applicable

- **Are all the source data underlying the results available to ensure full reproducibility?**

Yes

- **Are the conclusions drawn adequately supported by the results?**

Partly

Competing Interests

No competing interests were disclosed.

Reviewer Expertise

Political science

[1] <https://chainspect.app/dashboard>

Competing Interests: No competing interests were disclosed.

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